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# MICROMECHANICS OF THERMALLY INDUCED CRACKING IN THREE CRUSTAL ROCKS

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Abstract. Samples of Westerly granite, Frederick diabase, and Oak Hall limestone were thermally cracked at room pressure to various peak temperatures. Scanning electon microscopy (SEM) was performed on ion-thinned samples. The thermally induced crack density is dependent upon the temperature, thermal expansion mismatch, thermal expansion anisotropy, initial crack porosity, and grain size. The mode of propagation is different for grain boundary and intragranular cracks. Crack densities in the granite were quantified using stereological techniques. The thermally induced crack surface area per unit volume apparently has a quadratic dependence on the temperature increase, a physical interpretation for which can be formulated on the basis of energetic balance. Fracture mechanics models are developed to interpret thermal cracking. The predictions concerning thermal crack initiation temperature and crack propagation and arrest behavior agree well with observations for the granite and the diabase. The model predicts significant thermal cracking for the Oak Hall limestone, which contradicts the SEM observation. A possible explanation for this discrepancy is that the internal stresses due to thermal expansion anisotropy are relaxed by plastic flow in this relatively fine grained limestone.

## Introduction

Rocks are heated or cooled in many energy resources applications. Since rocks are polycrystalline aggregates with a dense population of microcracks and pores, thermal cracking results from the internal stress concentration induced by thermal expansion anisotropy (in the case of a monominerallic aggregate) or thermal expansion mismatch (in a multiphase aggregate). Such microcracking is coupled to that which results from thermal stresses induced by thermal gradients [Boley and Weiner, 1960] which may even arise in a perfectly homogeneous solid.

Laboratory studies have shown that appreciable microcracking in crustal rocks can be induced by thermal expansion anisotropy and mismatch even in the absence of thermal gradients [e.g., Richter and Simmons, 1974] and under confining pressure [e.g Wong and Brace, 1979; Heard and Page, 1982; Bauer and Handin, 1983]. The thermally induced cracks can influence significantly both the mechanical and transport properties, including thermoelastic moduli [Simmons and Cooper, 1978; Bruner, 1979; Heard and Page, 1982], ultrasonic velocities [Johnson et al., 1978], seismic attenuation [Johnston and Toksoz, 1980; Clark et al., 1981], permeability [Bauer and Johnson, 1979], thermal conductivity [Durham and Abey, 1981], fracture toughness [Meredith and Atkinson, 1985] and acoustic emission [Johnson et al., 1978; Chen and Wang, 1980]. It has been suggested that thermal cracking can strongly influence the state of stress at shallow depths [Bruner, 1984; Sbar et al., 1984], the transport processes in hydrothermal systems of crystallizing magmas [e. g., Norton and Knight, 1977] and the origin of the pore structures in crustal rocks [Nur and Simmons, 1970]. A fundamental understanding of thermal cracking mechanisms is of importance to energy resources applications in geothermal energy recovery [e. g., Fehler, 1982; Pearson et al., 1983] and nuclear waste disposal [e. g., Wang et al., 1981].

In the interpretation of geophysical measurements in the above situations, one of the crucial questions which have to be addressed is: What are the effects of temperature changes on the pore structure and hence the thermophysical properties of crustal rocks? Advances have been made in the past two decades toward a basic understanding of the relationship between the pore structure and physical properties. It is now generally accepted that the thermophysical properties depend not just on the porosity but also on details of the geometry of the pore space [e. g., Walsh, 1965; O'Connell and Budiansky, 1974; Walsh and Brace, 1984].

It follows that one must make detailed observations of the evolution of the pore structure as a function of temperature in order to specifically analyze the effects of thermally induced cracking on various physical properties. The experimental studies cited above have focused on the empirical correlation of temperature with porosity and various physical properties. Previous microscopy studies of thermal crack statistics [Friedman and Johnson, 1978; Bauer and Johnson, 1979; Homand-Etienne and Troalan, 1984; Meredith and Atkinson, 1985] did not use ion-milled specimens for the scanning electron microscopy (SEM) observations. The omission of ion-milling in SEM specimen preparation may significantly bias the interpretation of microcrack morphology and statistics [Brace et al., 1972; Richter and Simmons, 1977; Kranz, 1983]. To our knowledge, there has been no unambiguous measurement of the microcrack statistics as a function of temperature in crustal rocks.

It is usually suggested that thermal cracking in a quartz-bearing rock results from the anomalously high volumetric thermal expansion coefficient of quartz [e. g., Simmons and Cooper, 1978] and, similarly, that in a calcitic rock is due to the anomalous thermal expansion anisotropy of calcite [e. g., Rosengren and Jaeger, 1968]. Of course, both thermal expansion mismatch and anisotropy contribute to thermal cracking in a

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TABLE 1. Description and Thermal History of Rocks Studied

| Rock | Sample and Maximum Temperature Subjected to, °C | | Modal Analysis,* % | Porosity,* % | Grain Size,* mm |
|---|---|---|---|---|---|
| Westerly granite, Rhode Island | WGT0 | room | 27.5 quartz | 0.20 crack | 0.75 |
| | WGT1 | 100 | 35.4 microcline | 0.70 pore | |
| | WGT2 | 165 | 31.4 $An_{17}$ | | |
| | WGT3 | 250 | 4.9 mica | | |
| | WGT4 | 365 | | | |
| | WGT5 | 500 | | | |
| | WGT6 | 620 | | | |
| Frederick diabase, Maryland | FDT0 | room | 48 $An_{67}$ | 0 crack | 0.175 |
| | FDT1 | 345 | 49 Augite-hypersthene | 0.10 pore | |
| | FDT2 | 500 | | | |
| | FDT3 | 745 | 1 mica | | |
| | FDT4 | 1030 | | | |
| Oakhall limestone, Pennsylvania | LST0 | room | 99 calcite | 0 crack | 0.075 |
| | LST1 | 300 | | 0.20 pore | |
| | LST2 | 500 | | | |

*Data are from Brace et al, [1965]

polycrystalline aggregate composed of several anisotropic minerals. It should be useful to have a quantitative assessment of the relative contribution of the mismatch and anisotropy effects. It has been implicitly assumed in some studies that thermal cracking occurs principally along grain boundaries. The validity of this assumption is important, as whether or not the cracks occur along grain boundaries or intragranularly has significant implications on the aspect ratio distribution and the connectivity of the pore space, which in turn affects the thermomechanical properties of the rock.

In the present study, we thermally cracked suites of three petrologically distinct rocks at room pressure and made detailed, systematic observations on ion-milled sections of the thermally cracked samples using the scanning electron microscope. Quantitative determination of microcrack statistics were made for one of the rock types (Westerly granite). We have developed fracture mechanics models to interpret our SEM observations. The approach is analogous to that of recent work on thermal cracking in ceramics presented by Evans and coworkers [Evans, 1974, 1978; Evans and Clarke, 1980].

## Experimental Procedures

### Sample Preparation and Electron Microscopy

Cylindrical cores of each rock (18 mm diameter and 10 mm length) were dried in vacuo at room temperature for a minimum of 12 hours and then heated to various peak temperatures (Table 1). Heating and cooling rates were less than 2°C/min and temperature was monitored with a chromel-alumel thermocouple touching the base of the specimen. A hollow pyrophyllite core was placed between the specimen and walls of the furnace to minimize convection. Temperature gradients measured along the axis of a dummy pyrophyllite sample were less than 3°C/cm.

Polished crack sections (100 μm thick) were prepared perpendicular to the axis of each sample and were ion-milled. Samples were then sputter-coated with 200 Å of gold-palladium alloy and examined with an ISI Supra IIIA SEM which, in addition to the standard secondary electron detector, was equipped with a Robinson backscatter unit. An energy dispersive X ray (EDX) analyzer was available for compositional analysis. We could conveniently distinguish grain boundary and intragranular cracks in Westerly granite using the backscattered electron yield since this imaging mode is sensitive to compositional differences.

### Rock Types

We selected for study three petrologically distinct rocks: Westerly granite, Frederick diabase, and Oak Hall limestone (see Table 1 for petrographic analyses). Both Westerly granite and Frederick diabase have been well studied in experimental rock mechanics, and hence measurements of various physical properties were readily available from the literature. To study microcracking due to the anomalously high thermal expansion anisotropy in calcite, we considered the Oak Hall limestone. Both the aggregate compressibility [Brace, 1965] and thermal expansion [Wong and Brace, 1979] of this carbonate rock are relatively isotropic.

### Quantitative Stereology

The thermally induced crack surface area in Westerly granite was characterized quantitatively using stereological techniques [Underwood, 1970]. Essentially, the method consists of simply counting the number of crack intersections with a parallel array of equally spaced test lines. The data were obtained directly from the SEM, and the line intercept process was simulated by marking a straight line on the SEM monitor and counting the intercepts as the sample was displaced in a fixed

TABLE 2. Crack Density Data for Westerly Granite

| Sample | $P_L$* Grain Boundary, /mm | $P_L$ Intragranular*, /mm | $P_L$ Total**, /mm | $S_v$ Total$^{\Delta}$, $mm^2/mm^3$ |
|---|---|---|---|---|
| WGT0 | 2.11 ± 0.55 | 2.12 ± 0.86 | 4.23 ± 1.02 | 8.46 |
| WGT1 | 2.05 ± 0.63 | 2.33 ± 0.67 | 4.38 ± 0.85 | 8.76 |
| WGT2 | 2.64 ± 0.64 | 2.78 ± 0.92 | 5.42 ± 0.98 | 10.84 |
| WGT3 | 2.59 ± 0.66 | 2.65 ± 0.74 | 5.24 ± 1.08 | 10.48 |
| WGT4 | 2.73 ± 0.85 | 3.46 ± 0.85 | 6.20 ± 1.09 | 12.39 |
| WGT5 | 2.98 ± 0.77 | 4.73 ± 1.00 | 7.71 ± 1.37 | 15.42 |
| WGT6 | 2.77 ± 0.51 | 6.07 ± 1.33 | 8.84 ± 1.56 | 17.69 |

*The stereological data $P_L$ (mean cracks per unit length) were obtained directly from the SEM. The line intercept process was simulated by marking a straight line on the SEM screen and counting the intercepts as the sample was displaced in a fixed direction. Our test grid consisted of 32 parallel lines of length 4.0 mm and spaced by 0.125 mm. The standard deviation reflects the mineralogical variability of the rock when sampled at intervals of 0.125 mm.

**$P_L$ Total was obtained by adding $P_L$GB and $P_L$IG for each traverse.

$^{\Delta}S_v$ (crack surface area per unit volume) was obtained using equation (1).

direction. Our test grid consisted of 32 parallel lines of length 4.0 mm and spaced by 0.125 mm. Total test area was thus 16 $mm^2$. In all, more than 30 grains were covered. Samples were scanned at 1 kX and magnifications up to 20 kX were used to check ambiguous features. The criterion for a crack was simply that it has a visible aperture (at 1 kX) and an aspect ratio greater than about 1:10. This restriction served to screen from count the numerous, almost equidimensional pores naturally present in the plagioclase. Grain boundary and intragranular cracks were counted separately. Data for both grain boundary and intragranular cracks were then statistically analyzed and interpreted in terms of mean cracks per unit length ($P_L$). Rather than being a measure of experimental error, the standard deviation of such a measurement, as presented in Table 2, simply reflects the mineralogical variability of the rock when sampled at intervals of 0.125 mm.

Using arguments of geometric probability theory [Underwood, 1970], it may be shown that for an isotropic solid, this measurement made in the plane may be interpreted in terms of the three dimensional structure by the relation

$$S_V = 2P_L \qquad (1)$$

where $S_V$ is the crack surface area per unit volume. The validity of this equation does not depend on either the shape, connectivity, or any other geometric characteristics of the pore space. The only essential requirement is that the microcracks are randomly oriented on a macroscopic scale. The thermomechanical behavior of both the unheated and heated samples of Westerly granite are relatively isotropic. As mentioned above, our measurements covered about 30 grains and hence the use of equation (1) is probably justified. Also, observe in (1) that we count one crack as one surface.

In order to aid the interpretation of the thermal crack density data, additional stereological measurements similar to those described above were made of the grain boundary area per unit volume in Westerly granite. Measurements were made on a polarizing microscope on a standard thin section of sample WGT0. Grain boundary intersections with an array of test lines were counted, and in each case the mineralogical nature of the contact was noted (e.g., quartz/feldspar, feldspar/feldspar, quartz/biotite, etc.). Plagioclase feldspar and microcline cannot be readily distinguished with a polarizing microscope and hence both are referred to as simply feldspar. The test array consisted of 12 parallel lines 10 mm long and spaced by 1 mm, yielding a total test area of 120 $mm^2$. Data were then statisticlly analyzed and interpreted in terms of mean grain boundaries per unit length ($P_L$). The standard deviation of this measurement (Table 3) reflects the natural distribution of grain sizes in the rock. Assuming isotropy, equation (1) may then be used to interpret the data in terms of grain boundary area per unit volume.

## Electron Microscopy Observations

### Westerly Granite

Our stereological data for Westerly granite is summarized in Tables 2 and 3 and Figures 1 and 2. Detailed studies of the microcrack morphology [Sprunt and Brace, 1974] and statistics [Hadley, 1976] of virgin Westerly granite have been conducted previously and will therefore not be discussed here.

Grain boundary cracking. Thermally induced grain boundary cracking is concentrated in a very narrow temperature range. The onset of significant cracking occurs in the 100°-165°C interval, and a substantial proportion is achieved upon heating to 250°C (Figure 2a). Quartz/feldspar grain boundaries are observed to crack more readily than microcline/plagioclase boundaries, and in fact, any grain boundary remaining sealed upon heating to 360°C invariably involves two feldspars. Essentially, all grain boundaries are cracked upon heating to 500°C.

Table 3. Grain Boundary Data for Westerly Granite

| Grain Boundary | $P_{L'1}^{*}$ $mm^{-1}$ | $S_v^{**}$ $mm^2/mm^3$ | Percent, % |
|---|---|---|---|
| Quartz/feldspar | 1.35 | 2.70 ± 0.77 | 46 |
| Feldspar/feldspar | 0.79 | 1.58 ± 0.69 | 27 |
| Quartz/quartz | 0.21 | 0.43 ± 0.27 | 7 |
| Biotite/quartz or feldspar | 0.57 | 1.14 ± 0.37 | 20 |
| All | 2.92 | 5.87 ± 0.85 | 100 |

*$P_L$ was measured with a polarizing microscope by counting the grain boundary intercepts as the sample was displaced in a fixed direction. Our test grid consisted of 12 parallel lines of length 10 mm and spaced by 1 mm. The standard deviation represents the distribution in grain sizes in the rock.

**$S_v$ was obtained from $P_L$ using equation (1).

Grain boundary cracks are observed to initiate primarily at triple junctions (locations where three grains come into contact). Once initiated, grain boundary cracks tend to grow unimpeded until encountering a juncture in the grain boundary involving a sharp change in geometry, at which time the crack will either continue growth or stop entirely (Figure 3). These junctures serve as sites of renewed growth and as secondary sites of crack initiation. Discontinuities such as pores along or very close to the grain boundary have a similar effect, either retarding or promoting growth.

The value of $P_L$ for grain boundary cracks in the samples heated to 360°C and higher is about 2.8 $mm^{-1}$ (Figure 2a). Our independent measurement on sample WGTO of $P_L$ for grain boundaries is 2.9 $mm^{-1}$ (Table 3). This reinforces our qualitative observation discussed above that essentially all grain boundaries are cracked in the higher-temperature samples.

Intragranular cracking. During the critical temperature interval discussed above for grain boundary cracking (i.e., until 250°C), the rates of intragranular and grain boundary cracking are comparable (Figure 2). However, after heating to temperatures above 250°C (after which most of the grain boundaries have already cracked), there is a drastic increase in the rate of intragranular cracking (Figure 2b). Throughout this stage of dominantly intragranular cracking, three concurrent processes augment crack development: initiation and growth of intragranular cracks, continued incremental propagation of cracks, and widening of grain boundary crack apertures.

Intragranular cracks propagate preferentially in the feldspars, with microcline slightly favored over plagioclase.

Crack development is as outlined below:

1. There is initiation of an intragranular crack at a grain boundary and propagation in a direction roughly perpendicular to the boundary, gradually tapering inward until termination leaving a sharp tip, generally after propagation of the order of 0.20 mm or less (Figure 4).

2. Then there is incremental growth of existing intragranular cracks with planes of cleavage particularly favoring propagation (Figure 5). Maximum length is generally of the order of 0.40 mm (usually achieved only in the 500°C and 620°C samples).

3. Finally there is initiation and propagation of cracks from existing intragranular cracks in a direction roughly perpendicular to existing crack. Generally these terminate after propagation on the order of 0.20 mm or less, frequently upon encountering another crack.

Upon heating to 500° and 620°C, well-developed crack arrays exist in both feldspars (Figure 6). A single plagioclase grain is typically dissected into numerous roughly equidimensional blocks, while microcline is split-up into more rectangular sections.

Intragranular cracking in biotite and quartz is less well developed than that observed in the feldspars, with quartz showing the least of all. Intragranular cracks in quartz are only observed in those instances in which quartz is bounded by either biotite or another quartz grain. That is, an intragranular crack initiating at a quartz/feldspar contact is usually observed to propagate into the feldspar. Modes of propagation are similar to that discussed above for the feldspars with the exception that cleavage cracking is not evident. Quartz subgrains are usually more spherical and irregular in shape than feldspar subgrains, owing to the absence of a well-developed cleavage.

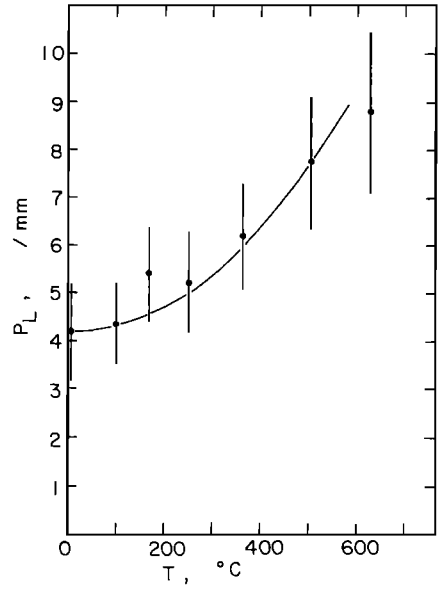



Fig. 1. The stereological parameter $P_L$ plotted as a function of temperature for Westerly granite. The solid line represents the theoretical curve $S_V - S_V^0 = 3.08 \times 10^{-5}$ ($mm^2/mm^3/°C$) $(\Delta T)^2$ (equation (9b)). The bar indicates one standard deviation which reflects the mineralogical variability of the rock when sampled at intervals of 0.125 mm.

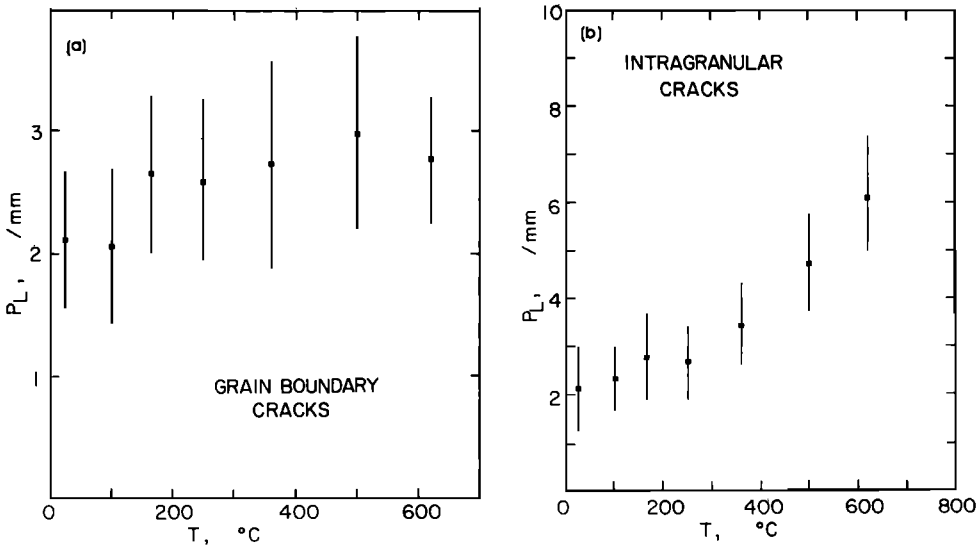



Fig. 2. $P_L$ plotted as a function of temperature for Westerly granite: (a) grain boundary cracks and (b) intragranular cracks.

Cracking in biotite appears to be controlled by the orientation of the (001) plane to neighboring grains. Both uncracked and highly cracked grains are observed in the higher-temperature samples. Crack arrays typically consist of a series of 10 or 15 individual cracks which almost always propagate in parallel directions to one another (Figures 7). As with the feldspars, cracks propagate in a direction perpendicular to the grain boundary at which they initiated. However, in biotite, intragranular cracks are frequently observed to traverse the entire grain straight across.

## Frederick Diabase

The crack porosity of this rock is zero [Brace, 1965]. The unheated sample exhibits few open cracks. Also present are healed/sealed microcracks which occur intragranularly in both pyroxene and plagioclase. Although the two processes of crack healing and sealing are distinct [Richter and Simmons, 1977], we use the term ''healed/sealed'' here since the EDX analyses were generally not sufficiently unambiguous to warrant the usage of one term over the other.

The temperature for the onset of thermal cracking in the diabase is significantly higher than that in the granite. Appreciable microcracking in the diabase is only observed in samples heated to temperatures above 500°C (Figure 8). Furthermore, the thermal crack density is relatively low for both the 500° and 745°C samples. A significant increase in the microcrack density is observed after heating to 1040°C (Figure 9).

It is difficult to assess quantitatively the relative contribution of grain boundary versus intragranular cracking in this rock. The observation of pyroxene/plagioclase grain boundaries was frequently obscured due to imaging effects associated with the large topographical relief which occurs along grain boundaries due to the large difference in ion-thinning rates for the two minerals. Intragranular cracking was observed in both pyroxene and plagioclase grains. Transgranular cracking was also common (Figure 9). Crack propagation was also observed to occur along previously healed/sealed crack surfaces.

Heating to 1040°C resulted in the formation of a partial melt, as evidenced by the presence of glassy textures, lobate grain boundaries, and other textural criteria. Melting appeared to nucleate around biotite crystals (which constitute less than 1% by volume). We suspect that the presence of this hydrous phase along with other possible sources of water (e. g., fluid inclusions, clay along grain boundaries, etc.) served to lower the melting point typically expected of a diabasic rock (roughly 1200°C). Interestingly, the partial melt appears to have migrated and filled many of the previously open thermal cracks (Figure 9). As indicated by EDX analysis, the material filling the cracks was iron-enriched, as would be expected of an early melt. (The bright color of the material filling some of the cracks is due to the high atomic number of iron.)

## Oak Hall Limestone

The initial crack porosity of the unstressed limestone is zero [Brace, 1965]. We observe no significant amount of cracking throughout the entire limestone series. The only thermal cracking which is observed occurs after heating to 500°C as grain boundary and intragranular cracks related to dolomite grains (Figure 10) which constitute less than 1% of the bulk rock (Table 1).

## Fracture Mechanics Analysis of Thermal Cracking

Our SEM observations and stereological data on Westerly granite indicate that there are fundamental differences between the two modes of thermal cracking: grain boundary and intragranular cracking. The former seemed to have stopped after the samples were heated to a relatively low

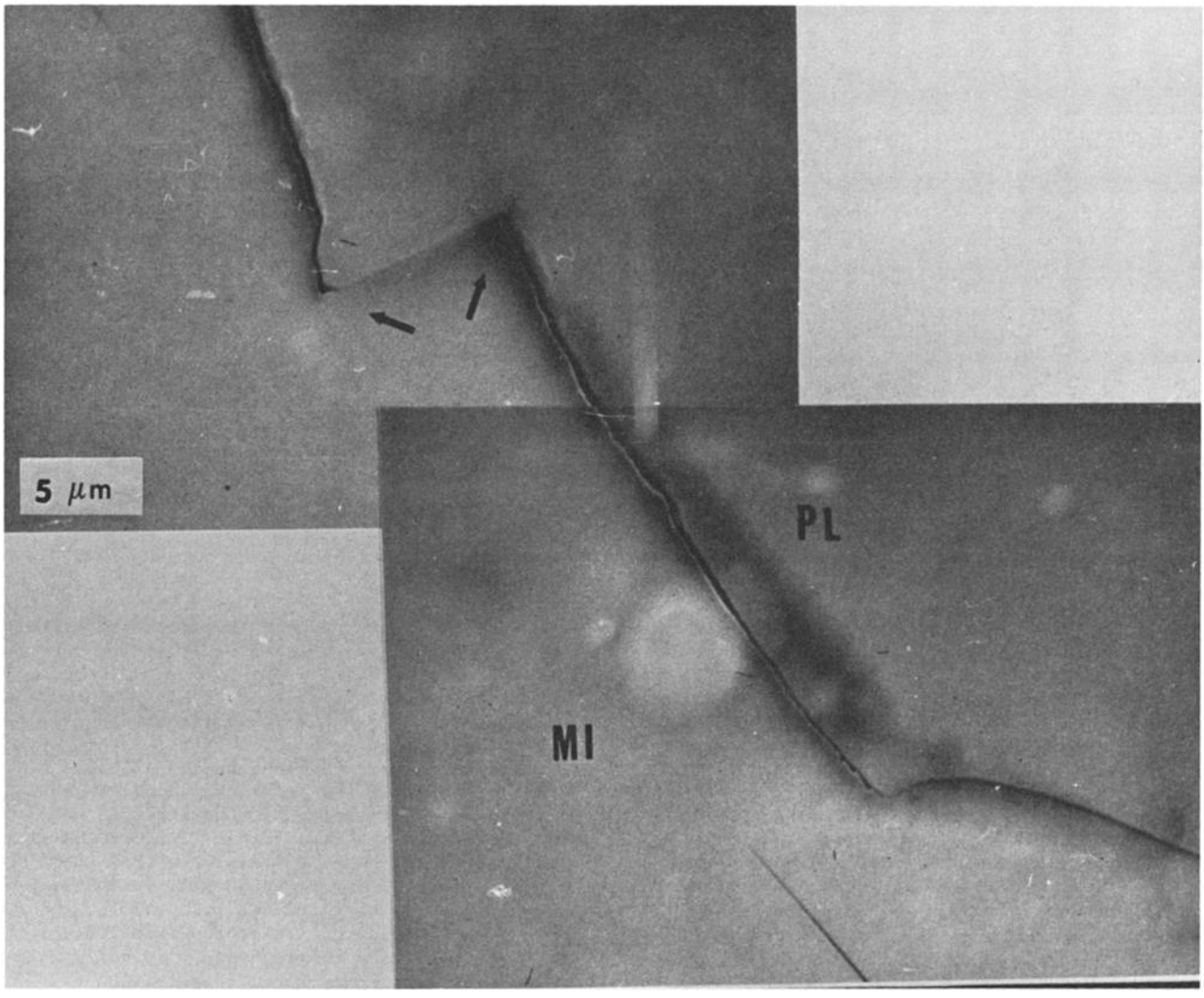



Fig. 3. Crack propagation along a plagioclase/microcline grain boundary in WGT2. The crack was arrested upon encountering the sharp corners along the boundary. (This and subsequent SEM micrographs are backscattered electron images.)

temperature of about 300°C (Figure 2a), at which point the thermally induced internal stresses have opened up a significant proportion of the grain boundaries. Furthermore, the grain boundary geometry has strong control over the propagation and arrest behavior of grain boundary cracking (Figure 3). On the other hand, intragranular cracking occurred continuously throughout the whole temperature range that we considered (Figure 2b). The rate of cracking accelerated when the temperature exceeded 300°C or so, but crack coalescence (such as what is generally observed in highly stressed samples, [e. g., Kranz, 1979; Wong, 1982b]) is not appreciable even for the 620°C sample (Figure 6).

Rocks are polycrystalline aggregates. The thermoelastic properties of each of the phases are anisotropic and have to be characterized by a fourth-rank tensor (elastic modulus) together with a second-rank tensor (thermal expansion coefficient). It is impossible to perform an exact calculation for the internal stress field within such an aggregate when it is subjected to a homogeneous temperature increase. Any mathematically tractable model for the micromechanical processes would involve certain idealizations with regard to both the grain geometry and the themoelastic moduli heterogeneity.

An inclusion model is commonly adopted for the analysis of thermal cracking in geologic [Johnson et al., 1978; Wong and Brace, 1979; Bruner, 1984] and ceramic [Evans, 1974; 1978; Clarke 1980; Davidge, 1981] materials. Solutions for the thermally induced internal stresses in the vicinity of an inclusion embedded within a matrix are sought, and then fracture mechanics concepts are applied to predict the thermal crack propagation and arrest behavior.

Most previous studies [Evans, 1974; Johnson et al., 1978; Wang and Heard, 1985] have used a spherical inclusion model for mathematical convenience. There are no sharp corners on a spherical surface. Our SEM observations show that the geometric junctures exert considerable control on grain boundary cracking. To take this into account, we performed a two-dimensional

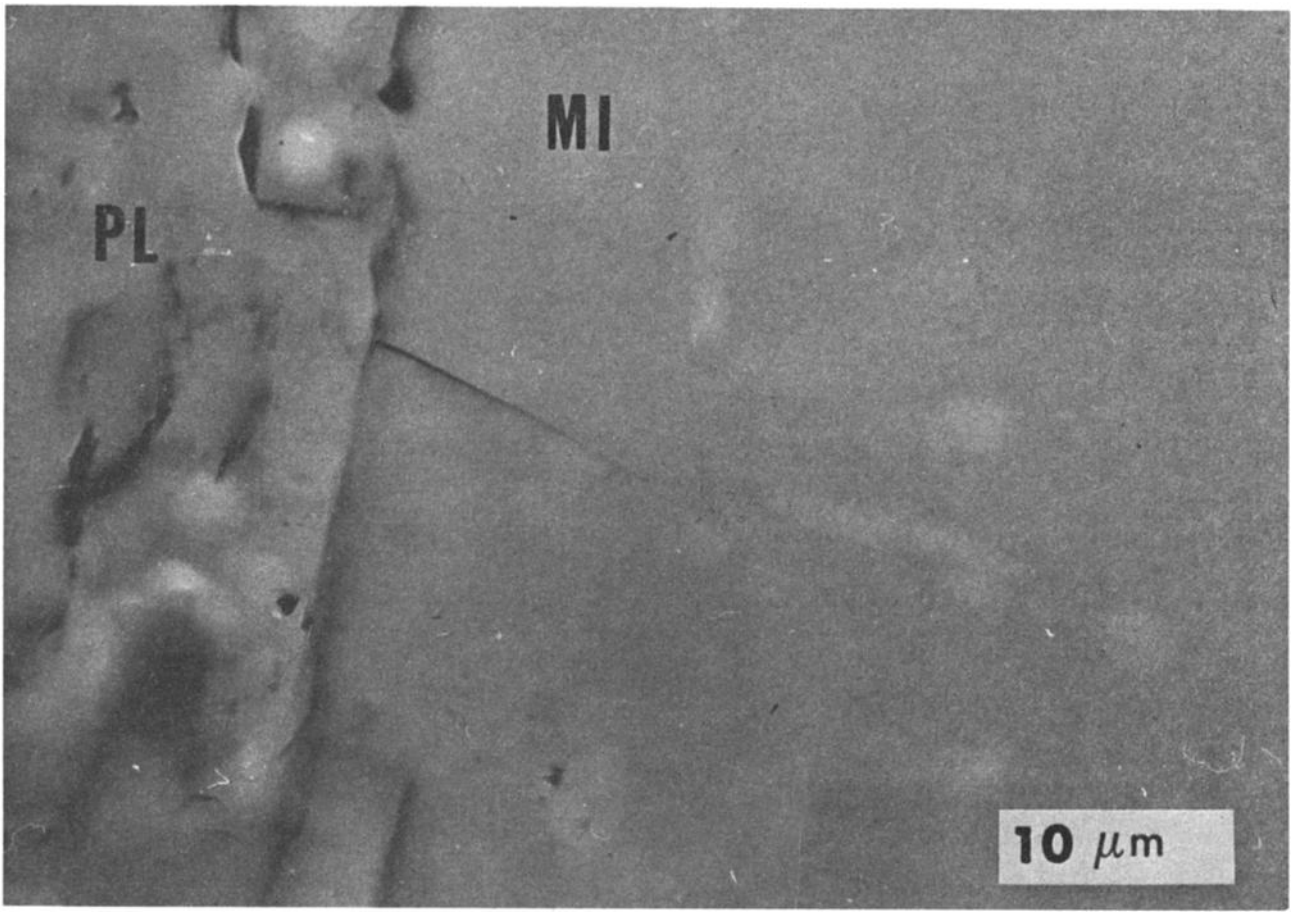



Fig. 4. Intragranular crack propagation in microcline in WGT1. The crack has propagated in a direction roughly perpendicular to the microcline/plagioclase grain boundary and was arrested after propagating for less than 20 μm. Note the sharp-tipped character of the crack.

calculation for a square inclusion. More complicated geometry can be analyzed [e. g., Evans, 1978; Bruner, 1984], but we believe that this relatively simple model (first suggested by Clarke [1980]) already captures most of the important features.

We obtained analytic solutions for the stress fields associated with a square inclusion (in a two-dimensional matrix). The stress intensity factor for the tensile mode ($K_1$) of a flaw located in proximity to the inclusion was then evaluated by the use of Green's function [Evans, 1978; Cartwright and Rooke, 1979]. The key conclusions will be summarized below, and the mathematical details are presented in Appendix 1.

Thermal Expansion Mismatch and Anisotropy

We consider the effects of a homogeneous temperature increase on the internal stress field of an anisotropic inclusion (with thermoelastic moduli appropriate for each of the constituent minerals) embedded within an effective medium (with thermoelastic moduli of the polycrystalline aggregate). As discussed above, all three rocks studied in this project are relatively isotropic, and hence the thermoelastic moduli of the matrix will be assumed to be isotropic. The internal stress field is expected to be a function of the temperature, the inclusion geometry, the difference in thermal expansion coefficient and the difference in elastic moduli. The contribution from the heterogeneity in elastic moduli to the internal stresses is expected to be relatively insignificant in comparison to the contribution from thermal expansion heterogeneity in both geologic [Wong and Brace, 1979; Bruner, 1984] and ceramic [Evans, 1978; Clarke, 1980] materials. Following the approach of such previous work, we will assume the elastic moduli to be identical for the matrix and the inclusion. The Young's modulus is denoted by E and the Poisson's ratio by v.

If the thermal expansivity of the matrix is $\alpha^o\delta_{ij}$ and that for the inclusion is $\alpha^1_{ij}$, then the tensor characterizing the difference between the inclusion expansivity and the matrix expansivity can be separated into an isotropic component and a deviatoric component:

$$\Delta\alpha = \alpha^1_{ii}/3 - \alpha^o \qquad (2a)$$

$$\Delta\alpha'_{ij} = \alpha^1_{ij} - (\alpha^1_{kk}/3)\,\delta_{ij} \qquad (2b)$$

The isotropic component represented by (2a) represents the thermal expansion mismatch, whereas the deviatoric part in (2b) represents the thermal

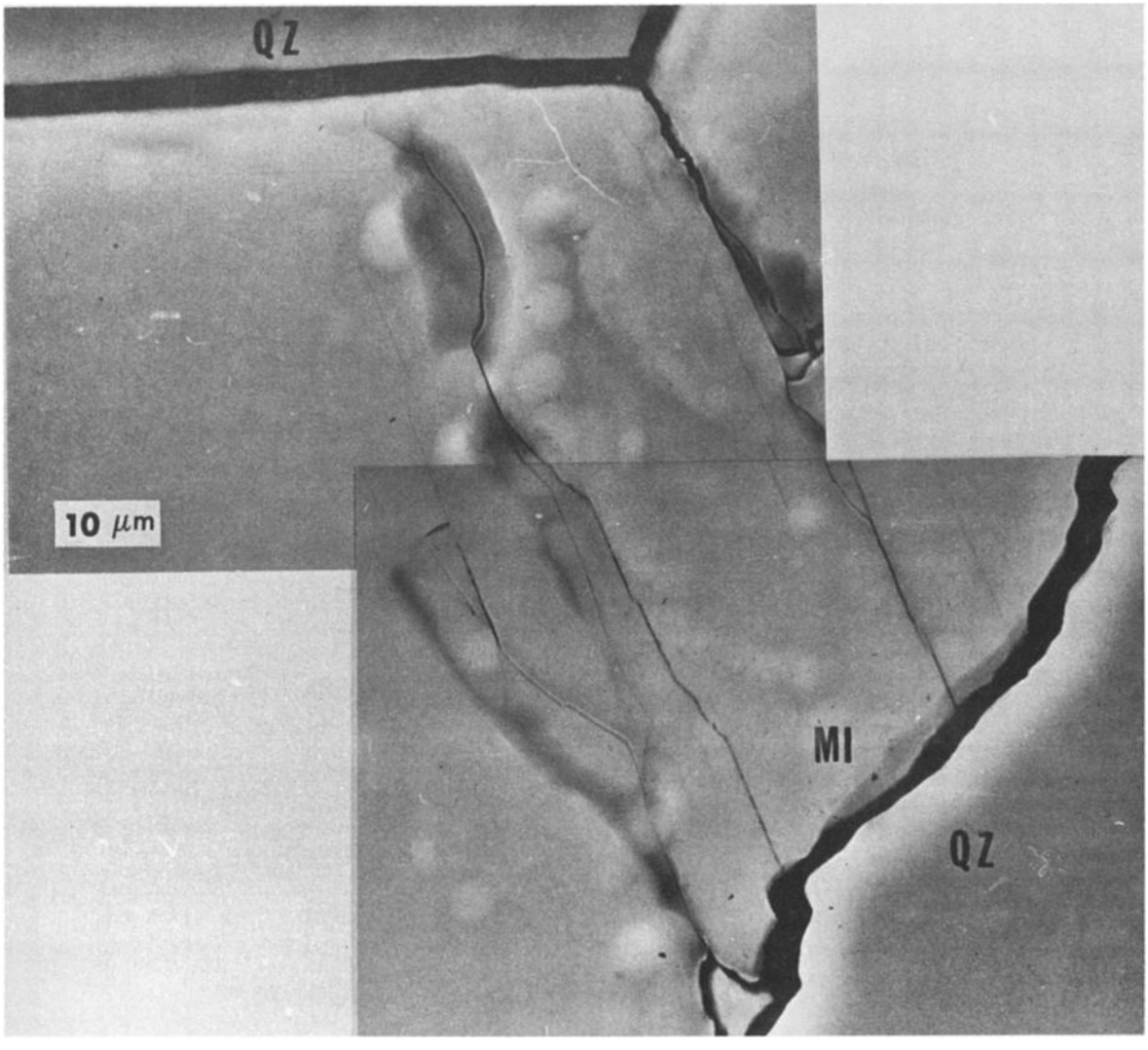



Fig. 5. Cracks propagating along what is probably a cleavage plane in microcline in WGT6. Note the extremely wide crack aperture along the quartz/microcline grain boundaries.

expansion anisotropy. We can solve for the individual contributions separately and then use the principle of linear superposition to calculate the total contribution from both thermal expansion mismatch and anisotropy. Tensile stresses are taken to be postive throughout this discussion.

Square Inclusion With Thermal Expansion Mismatch

Let the square inclusion have dimension 2L (Figure 11a). The general solution for the stress field outside the inclusion is given in Appendix 1. The critical location for an intragranular crack is expected to be in the diagonal direction as shown in Figure 11b. With reference to the coordinate system shown in Figure 11b, the normal stress along the x axis is given by

$$\sigma_{yy}(x) = \frac{E\Delta\alpha(T-To)}{2\pi(1-v)} \ln \left[\frac{(x+\sqrt{2}L)^2 + 2L^2}{x(x+2\sqrt{2}L)}\right] \quad (3)$$

Note that the stress has a singularity at each of the corners. It decays monotonically with distance from the corner. If the thermal expansion of the inclusion is higher than that of the matrix, then the stress is tensile. If the stress is sufficiently high, then intragranular cracks would initiate and propagate into the matrix. By symmetry, the shear stress along the x axis vanishes and hence only the mode I stress intensity factor needs to be considered.

Most rocks have a dense network of fine cracks even in the virgin state [e. g., Sprunt and Brace, 1974]. Thermally induced intragranular cracking is possible if the tensile stress intensity factor ($K_1$) at the crack tip exceeds a critical value ($K_{1c}$). As we discuss in Appendix 1, $K_1$ is given by [Paris and Sih, 1965; Evans, 1974; Cartwright and Rooke, 1979]

$$K_1 = \left(\frac{2}{\pi a}\right)^{1/2} \int_0^a \frac{\sigma_{yy}(x)\, x^{1/2}}{(a-x)^{1/2}}\, dx \quad (4)$$

for a two-dimensional crack of length a located along the x axis. One of the crack tips is at the origin, and the above value of $K_1$ is for the

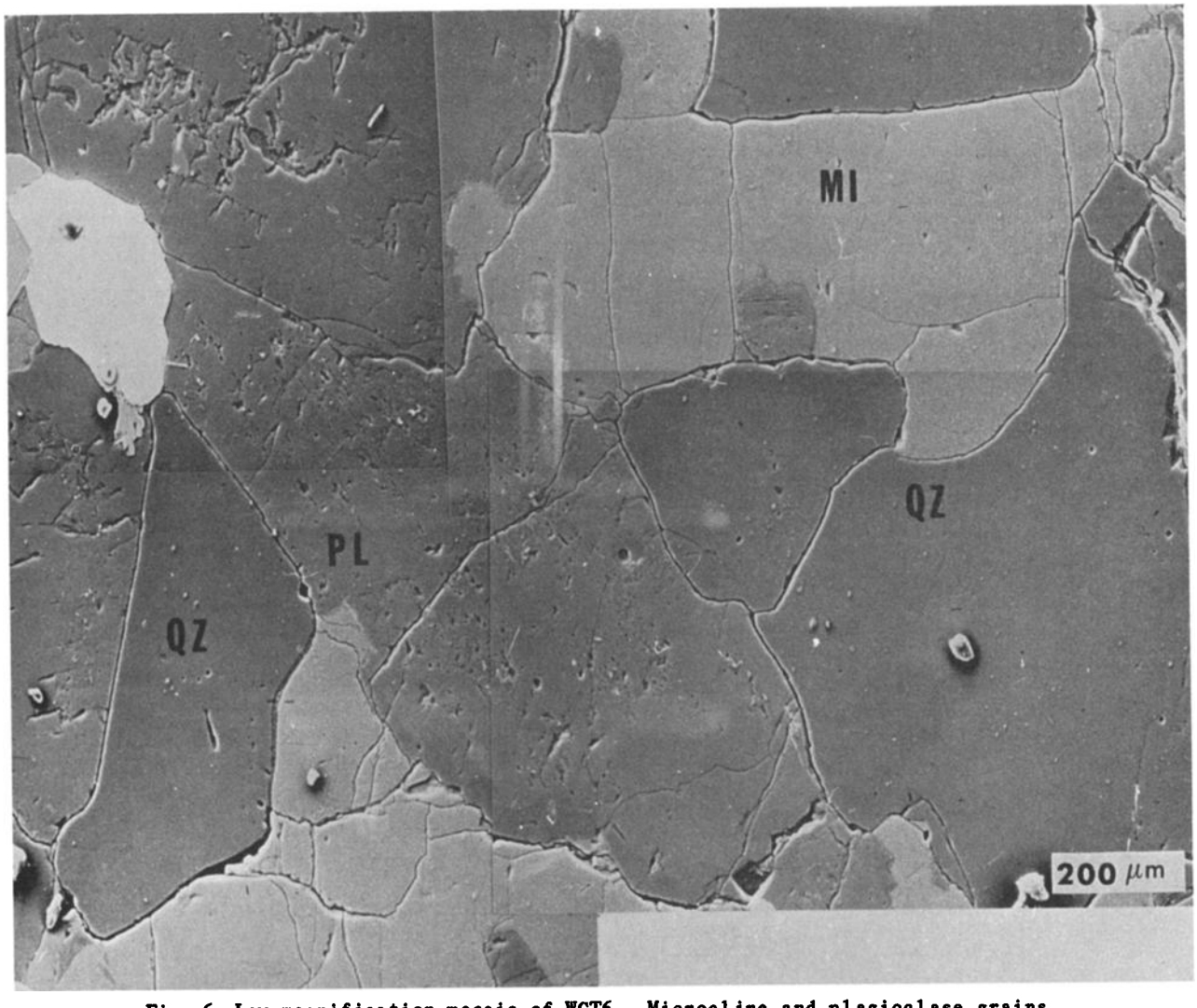



Fig. 6. Low magnification mosaic of WGT6. Microcline and plagioclase grains have a high density of cracks and quartz is basically uncracked. The very bright grain in the upper left is magnetite.

other tip at x=a. The normal stress given by equation (3) can be substituted into (4), and the stress intensity factor is evaluated accordingly by numerical integration. The numerical results are plotted in Figure 12a.

With reference to Figure 11c, the normal stress acting on the grain boundary is given by

$$\sigma_{yy}(x) = \frac{E\Delta\alpha(T-To)}{2\pi(1-\nu)} \left[\tan^{-1}\frac{x}{2L} + \tan^{-1}\left(1-\frac{x}{2L}\right) - \pi\right] \quad (5)$$

The term inside the bracket is negative, and therefore if the thermal expansion of the inclusion is less than that of the matrix, the stress is tensile possibly resulting in grain boundary cracking. The stress distribution is symmetric with respect to the midpoint of each side. It attains its maximum value at a corner and decreases monotonically toward the midpoint. The variation in normal stress is relatively small. Again (4) was used to evaluate $K_1$, and the numerical results are shown in Figure 12b.

The shear stress is not zero along the grain boundary (see equation (A2) in Appendix 1). Hence the crack will have a nonzero $K_2$. In a homogeneous material, such a mixed-mode crack will tend to deviate from its initial path along the grain boundary to take up a direction corresponding to pure mode I opening [Cotterell and Rice, 1980]. However, the fracture toughness for a grain boundary is probably much lower than that for the matrix, and hence once a crack is initiated on a grain boundary, it will prefer to propagate along the plane of weakness.

Figures 12a and 12b show that the two types of cracking are qualitatively different. $K_1$ for intragranular cracks peaks off and then decreases with increasing crack length. This implies that intragranular cracks induced by thermal expansion mismatch are expected to propagate over a relatively short distance before they are stabilized and arrested. Of course, further propagation is still possible if the temperature is increased. On the other hand, $K_1$ for grain boundary cracks increases monotonically with crack length which implies that as soon as it is

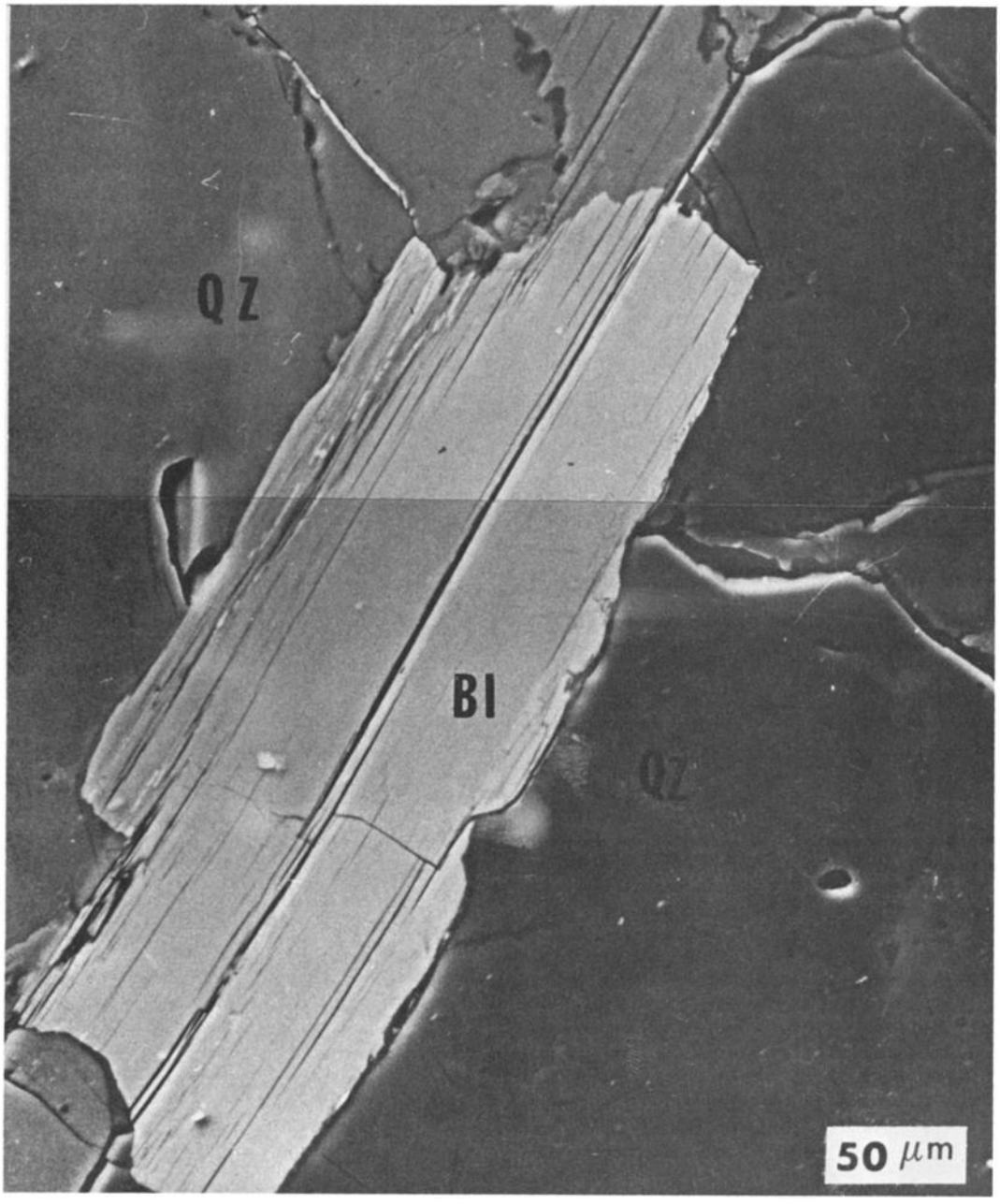



Fig. 7. Cracks along a cleavage plane in biotite in WGT6. The neighboring grain to the right and left is quartz.

initiated along a grain boundary, thermal cracking is unstable and the propagation will continue until the crack is arrested at a geometric juncture.

Square Inclusion With Thermal Expansion Anisotropy

In general, there are five independent components for the tensor $\Delta\alpha'_{ij}$ characterizing the thermal expansion anistoropy (equation (2b)). Complete solution for the stress field can be obtained for the general case [Chiu, 1977]. To highlight the effect, we will consider the configuration shown in Figure 13a (suggested by Clarke [1980]).

Two of the principal thermal expansion coefficients of the inclusion are parallel to the diagonals. One is higher and the second is lower than the matrix expansivity. The third principal thermal expansivity is assumed to be identical to that in the matrix. With respect to a coordinate system parallel to the diagonals, the thermal expansion anisotropy tensor is then given by $\Delta\alpha'_{11}=\Delta\alpha'$ and $\Delta\alpha'_{22}= -\Delta\alpha'$. For mathematical convenience, we used the coordinate system shown in Figure 13b. It is trivial to show that in this case, the only nonvanishing components of thermal expansion anisotropy tensor are $\Delta\alpha'_{12} = \Delta\alpha'_{21}= \Delta\alpha'$.

The stress acting normal to the diagonal direction at a location close to two of the four corners is expected to be tensile. We show in

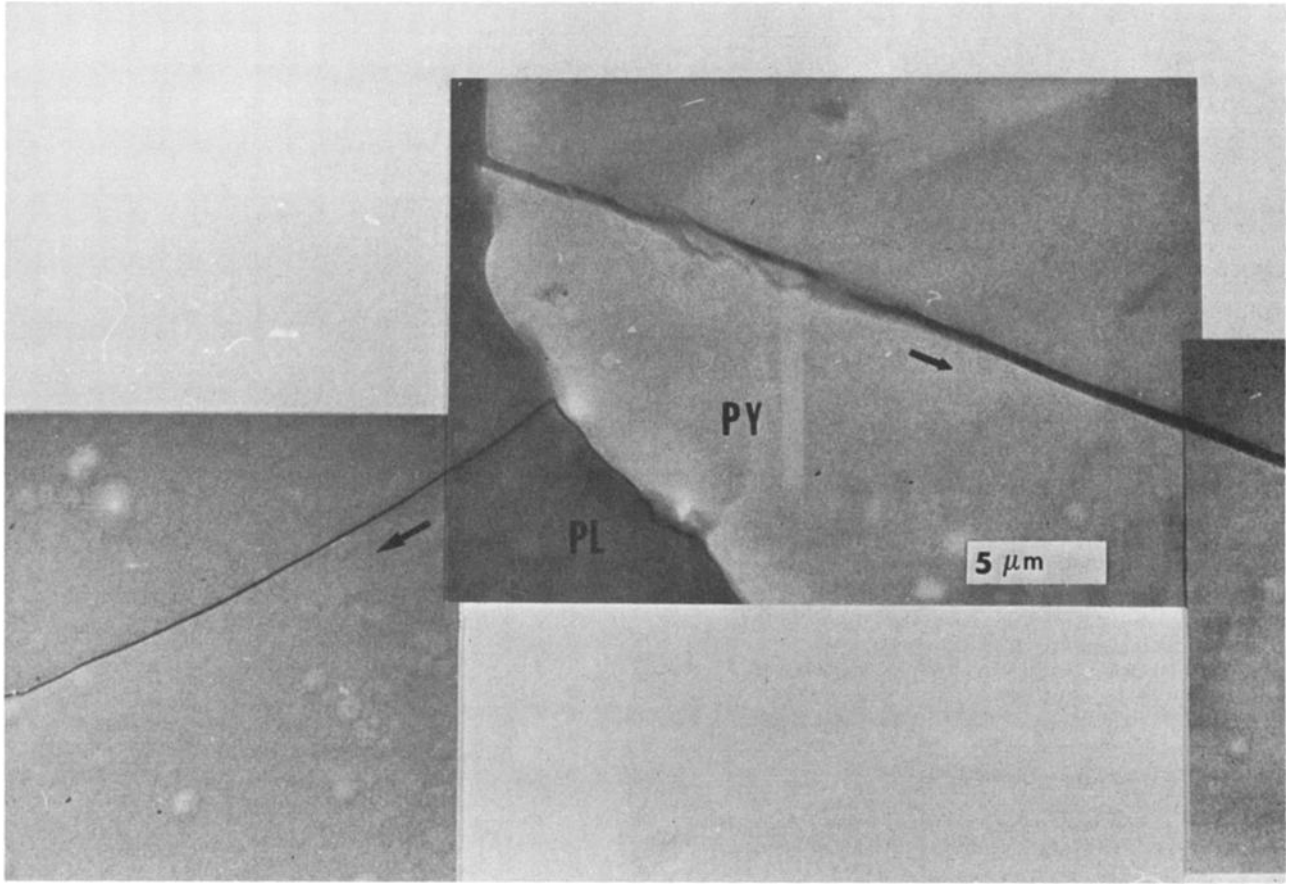



Fig. 8. The crack trending NE-SW initially propagated along the pyroxene/grain boundary and then changed direction and propagated either intragranularly in plagioclase or possibly along a plagioclase/plagioclase grain boundary. The crack was arrested farther on (not shown here). The intragranular crack in the pyroxene grain (oriented NW-SE) is not open but is filled with some material.

Appendix 1 that with respect to the coordinate system shown in Figure 13c, this stress is given by

$$\sigma_{yy}(x) = \frac{E\Delta\alpha'(T-To)}{2\pi(1-\nu^2)} \ln \left[\frac{(x+\sqrt{2}L)^2 + 2L^2}{x(x+2\sqrt{2}L)}\right] \quad (6)$$

Note that except for a factor of 1+v, the above expression is identical to that in (3) for the thermal expansion mismatch case. Hence for comparable magnitudes of thermal expansion mismatch and thermal expansion anisotropy, the propagation and arrest behavior of intragranular cracks would be quite similar. The numerical results for $K_1$ are shown in Figure 14a. The shear stress is zero and hence $K_2$ vanishes.

With reference to Figure 13d, the normal stress acting on a grain boundary is given by

$$\sigma_{yy}(x) = \frac{E\Delta\alpha'(T-To)}{2\pi(1-\nu^2)} \left[\frac{4L^2}{4L^2+(2L-x)^2} - \frac{4L^2}{4L^2+x^2}\right.$$

$$\left. + \ln\left(\frac{2L-x}{x}\right) - \frac{1}{2}\ln\left(\frac{4L^2+(2L-x)^2}{4L^2+x^2}\right)\right] \quad (7)$$

The above has a singularity at each of the corners. The stress distribution is antisymmetric with respect to the midpoint so that one half of the side is under tension, whereas the other half is under compression. The stress intensity factor for a microcrack initiating from one of the corners under tensile stress was evaluated by substituting (7) into (4) and the results are shown in Figure 14b. Note that the grain boundary cracking behavior for the thermal expansion anisotropy case is expected to be different from that for the thermal expansion mismatch case (Figure 12b). For the thermal expansion anisotropy case, $K_1$ actually decreases after reaching a peak as a function of crack length, indicating that grain boundary cracking induced by thermal expansion anisotropy is relatively stable and thermal crack arrest is generally expected.

## Discussion

### Relation of Surface Area of Thermal Cracks to Temperature

Our SEM observations on the thermal crack morphology are in qualitative agreement with most of the previous studies. However, our quantitative measurements of crack density in Westerly granite

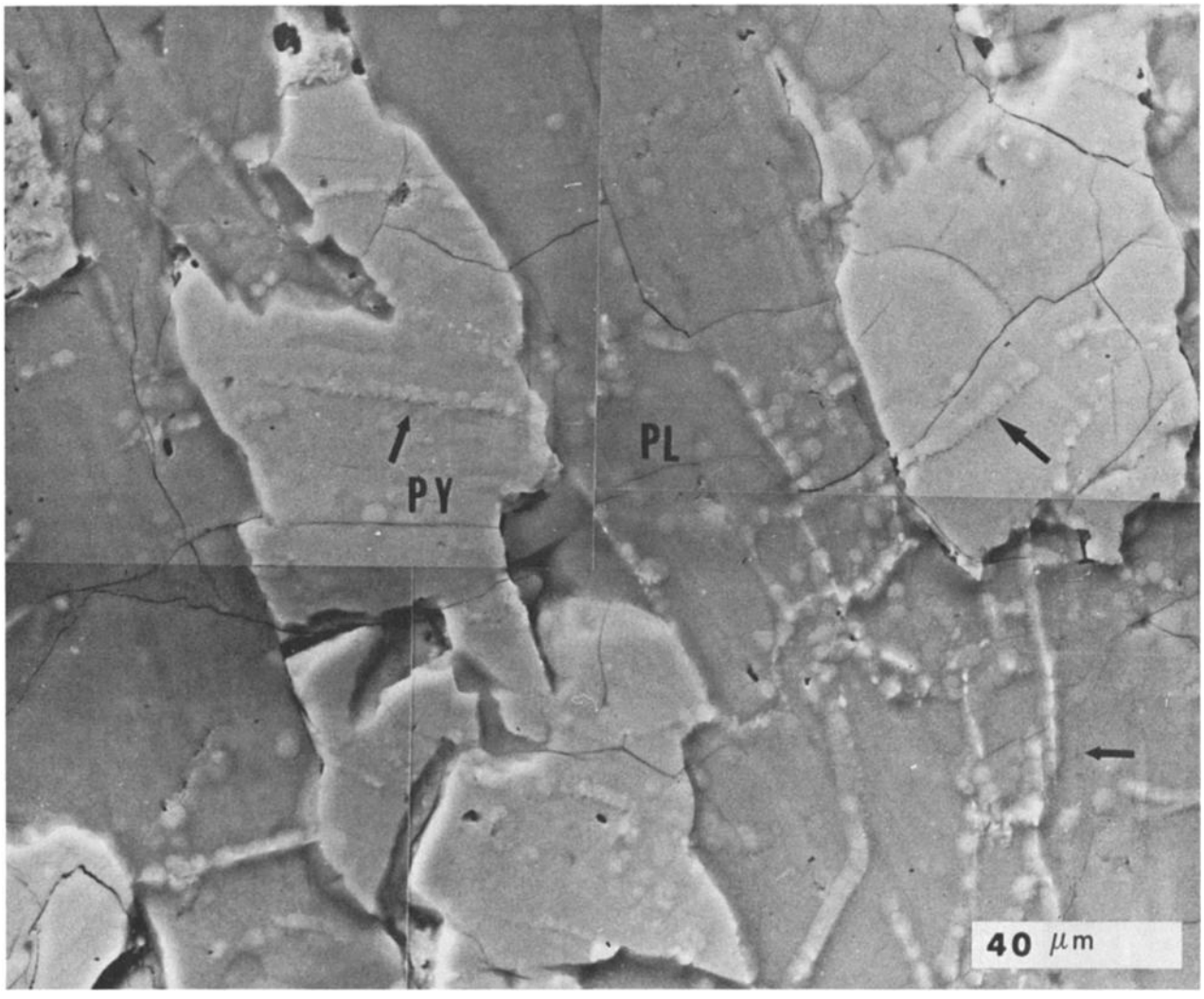



Fig. 9. Thermal cracks in a sample (FDT4) heated to 1040°C. The arrows indicate cracks which have been filled with partial melt. The brightness of the material filling the cracks in plagioclase is due to the high concentration of iron in the melt. Many of the cracks that appear to be open have actually been partially filled with the melt.

as a function of temperature are somewhat different from previous data obtained by Bauer and Johnson [1979] and Meredith and Atkinson [1985] using optical microscopy.

The data of Bauer and Johnson [1979] showed that basically all the grain boundaries were cracked for samples heated to temperatures above 500°C or so. Meredith and Atkinson's [1985] measurements of grain boundary crack density tend to be lower than those of Bauer and Johnson [1979]. However, both sets of data are consistent in that they both show a continuous increase in grain boundary cracking throughout the temperature ranges considered. On the other hand, our results (Figure 2) indicate that grain boundary cracking for Westerly granite occurs principally at relatively low temperatures below 300°C or so.

The intragranular crack density from optical microscopy studies are consistently lower than our values. Bauer and Johnson [1979] reported a value of about 1 for the ratio of intragranular $P_L$ to grain boundary $P_L$ in their 565°C sample, whereas our stereological data (Table 2) give a ratio of 1.6 for the 500°C sample and 2.2 for the 620°C sample. A possible explanation for such discrepancies is that the relatively short intragranular cracks with a small aperture can be better resolved by SEM observation on an ion-milled specimen. The aperture of the grain boundary cracks should also increase with temperature, and hence the cracks are more evident in the high-temperature samples when observed under an optical microscope.

SEM observations on the ion-milled specimens have provided enhanced resolution and elucidated the micromechanical processes. The fracture mechanics model discussed above was motivated by such observations, and quantitative comparisons with the observations will be discussed in a later section.

A number of recent models of the effects of cracks on the elastic moduli [Walsh and Grosenbaugh, 1979], seismic attenuation [Mavko, 1979; Stewart and Toksoz, 1983], electrical

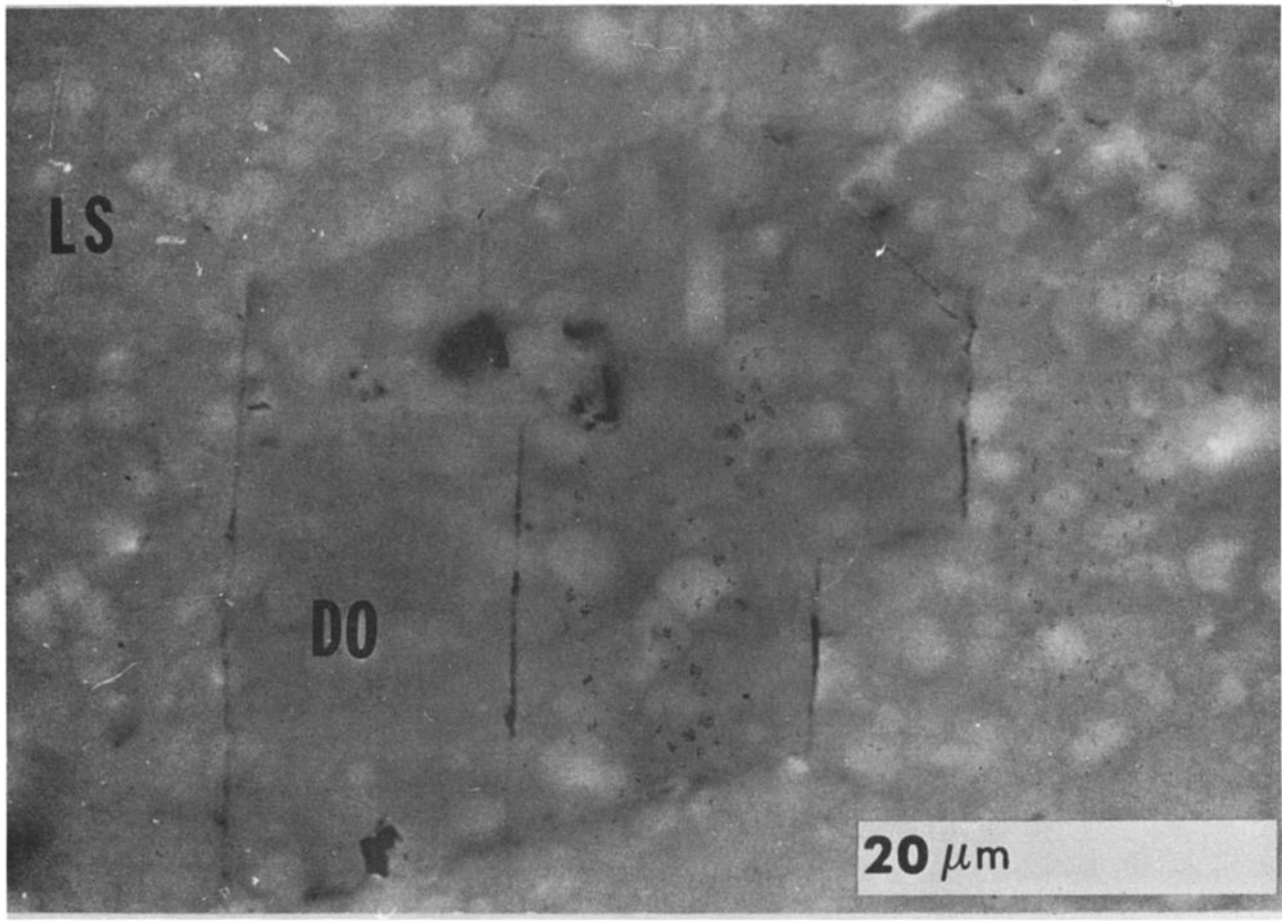



Fig. 10. Microcracking in LST2 is limited to that which occurs in connection with isolated dolomite grains. The matrix is calcite.

resistivity and permeability [Walsh, 1981; Walsh and Brace, 1984] are based on Greenwood and Williamson's [1966] asperity-contact theory. The overall crack surface per unit volume ( $S_V$ in equation (1)) is a key parameter in this type of model [Wong and Fredrich, 1984]. It should be useful to establish an empirical relation between $S_V$ and temperature.

We attempted to generalize a model proposed by Davidge and Green [1968] to interpret our data (Figure 1). Consider an isotropic spherical inclusion (of radius R) embedded in an isotropic matrix. If the elastic moduli are the same for the two and the mismatch in thermal expansion coefficients is $\Delta\alpha$, then a straightforward thermoelasticity calculation shows that the total strain energy (of the inclusion and the matrix together) is given by $[E/1-\nu]\ (\Delta\alpha)^2\ (T-T_o)^2 V$, where E and $\nu$ are the Young's modulus and Poisson's ratio, respectively, and V is the volume of the inclusion.

In their original analysis, Davidge and Green [1968] only considered grain boundary cracking. They showed that the matrix strain energy is highly localized: seven eighths of the energy is included in the matrix volume from r=R to r=2R. Hence one would expect the intragranular cracking also to be localized within this volume. Although the strain energy density is expected to have spatial variation, the average strain energy density within the sphere of radius 2R (and volume 8V) is approximately given by

$$W = 0.125\ [E/1-\nu]\ (\Delta\alpha)^2\ (T-T_o)^2 \qquad (8)$$

The occurrence of thermal cracking relaxes the internal stresses, and the thermoelastic strain energy is dissipated as surface energy for microcrack propagation. If the crack surface area per unit volume for an unheated sample is $S_V^o$, then an estimate of the fracture energy per unit volume [Wong, 1982b] is simply given by the product of the crack surface area $(S_V - S_V^o)$ and the fracture mechanics parameter $G_{1c}$ (critical energy release rate under mode 1 or tensile loading). Energetic balance for the thermal cracking process would then imply that

$$8\ (S_V - S_V^o)\ G_{1c} = [E/1-\nu]\ (\Delta\alpha)^2\ (T-T_o)^2 \qquad (9a)$$

The above can be rearranged to have the form

$$S_V - S_V^o = k\ (T-T_o)^2 \qquad (9b)$$

where $k = E\ (\Delta\alpha)^2 / [8(1-\nu)G_{1c}]$

If the temperature dependence of the thermoelastic moduli and the fracture energy parameter ($G_{1c}$) are such that k does not vary significantly with temperature, then a quadratic relation such

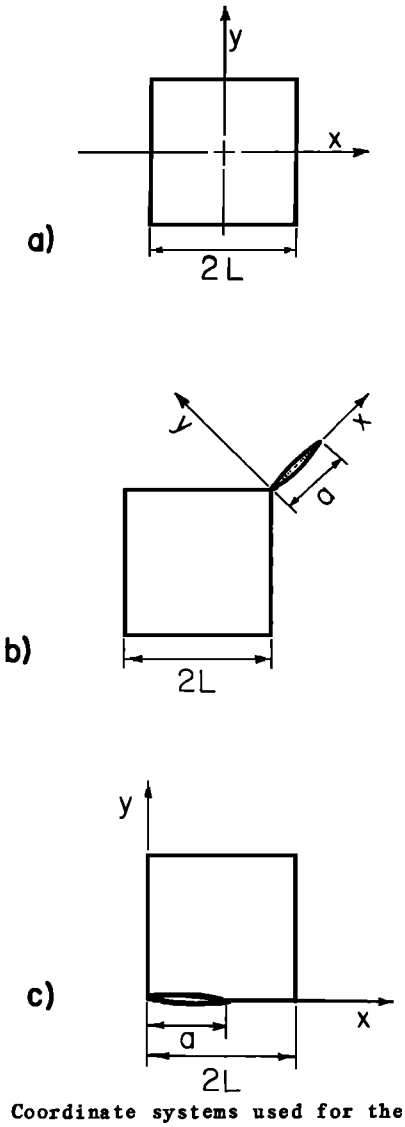


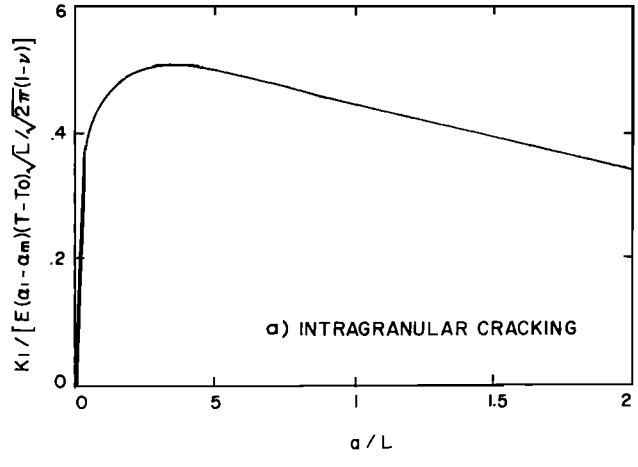

Fig. 11. Coordinate systems used for the analysis of thermal cracking due to thermal expansion mismatch in the square inclusion model: (a) corresponds to equation (A2), (b) equation (3), and (c) equation (5).

as that shown in Figure 1 is to be expected. A similar conclusion can be reached simply on the basis of dimensional analysis, but the above energetic balance calculation has the merit of highlighting the important roles of the various physical processes.

Representative values of E and v for Westerly granite under room condition are 56 GPa [Johnson et al., 1978] and 0.1 [Brace, 1971] respectively. An average value for $G_{1c}$ of the major phases of a granite is 7.5 J/m$^2$ [Atkinson, 1984]. By fitting a quadratic curve to our stereological data (Figure 1) and assuming isotropy (equation (1)), we determined k to be 3.08 X $10^{-5}$ mm$^2$/mm$^3$/°C and hence the average mismatch in thermal expansion coefficients as predicted by the spherical model is about 7 X $10^{-6}$/°C, a reasonable value for Westerly granite (see Appendix 2).

The quadratic curve is by no means the only one which fits our data. A number of assumptions have been made in deriving (9b). Quartz is expected to undergo the α-β transition at 573°C, and therefore an additional term ought to be included in the energetic balance calculation for samples thermally cracked to temperatures above this transition temperature. One would expect both the Young's modulus [Johnson et al., 1978] and the fracture energy [Atkinson, 1984] to decrease as temperature increases. On the other hand, the thermal expansion coefficent [Skinner, 1966; Roy et al., 1981] and possibly the thermal expansion mismatch (Δα) are expected to increase with temperature. The factor of 8 in (9a) should not be accepted as an exact value but rather as a geometric factor characterizing the spatial distribution of the internal strain energy.

Existing data are insufficient for a more accurate assessment of k. We have performed more involved calculations using Walsh's [1973] approach, which takes into account both the thermal expansion anisotropy and mismatch. The results are qualitatively the same, but we believe that there is no point in pursuing further refinement in such theoretical calculations in the absence of more data.



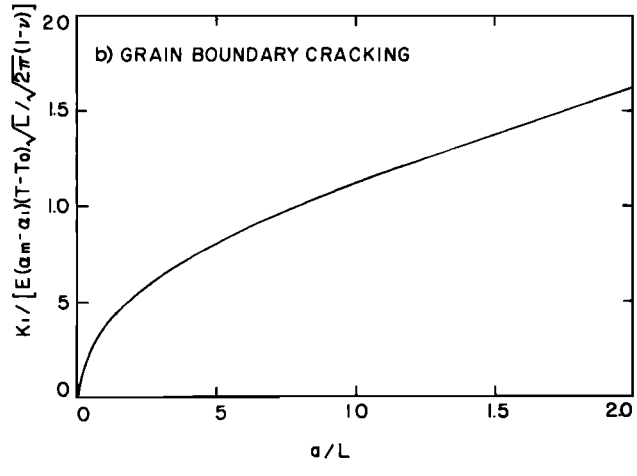




Fig. 12. Normalized stress intensity factor $K_1$ for cracking due to thermal expansion mismatch in the square inclusion model plotted as a function of the normalized preexisting crack length a: (a) intragranular and (b) grain boundary cracking.

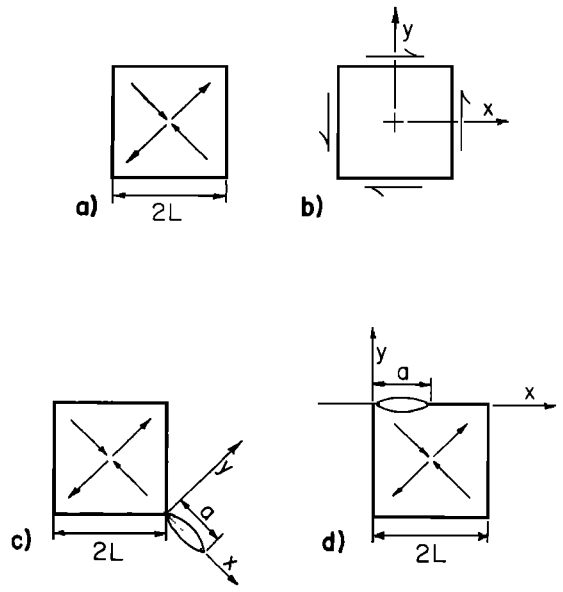



Fig. 13. Coordinate systems used for the analysis of thermal cracking due to thermal expansion anisotropy in the square inclusion model: (a) orientation of the principal thermal expansion coefficients (the third is assumed to be identical to the matrix), (b) corresponds to equation (A3), (c) equation (6), and (d) equation (7).

## Comparison of SEM Observations and Fracture Mechanics Model

It is important to keep in mind the limitations of the fracture mechanics models before a comparison of the theoretical results and SEM observations is made. Heterogeneity in thermoelastic properties of neighboring grains has been smoothed by the use of a homogeneous, effective medium. Heterogeneity in elastic moduli has also been neglected. A comparison between our results for a square inclusion and those for a spherical inclusion model [Evans, 1974; Ito et al., 1981] also highlights the sensitivity of the results to grain geometry.

The inclusion is assumed to be welded to the matrix in our model. The SEM observations showed that intragranular cracking can occur after extensive grain boundary cracking. If the coefficient of friction along the cracked grain boundaries is sufficiently high, our model should still be a good approximation. On the other hand, if the coefficient is relatively low, then the thermally induced internal stresses in the matrix will be reduced [Ghahremani, 1980]. The propagation of an intragranular crack probably would still be stable, but the threshold temperature for crack propagation is expected to be higher than that predicted by our current model.

Our model is deterministic in that if the initial crack length and the critical stress intensity factor are known, then the threshold temperature for crack propagation and the crack arrest behavior can be predicted. In a polycrystalline aggregate it may be more realistic to treat the microcrack dimension and fracture toughness distribution as statistical in nature. In this sense, the fracture mechanics models should be considered to be only semiquantitative. However, they provide insights useful to the interpretation of the micromechanical processes, particularly with regard to the evaluation of the relative contributions of thermal expansion mismatch and anisotropy and to a basic understanding of the qualitative difference between thermally induced grain boundary and intragranular cracking.

Our analyses have been influenced strongly by recent materials science researches, and the approach is different in a number of ways from some of the related studies in the rock mechanics literature. A spherical inclusion was considered by Johnson et al. [1978] and Wang and Heard [1985]. Our results show that geometric junctures on the grain boundary play an important role and should be included in the model. Fracture mechanics concepts were not used in such studies. The important roles of grain size and initial flaw distribution in influencing the thermal cracking behavior could not be considered solely on the basis of stress analysis.

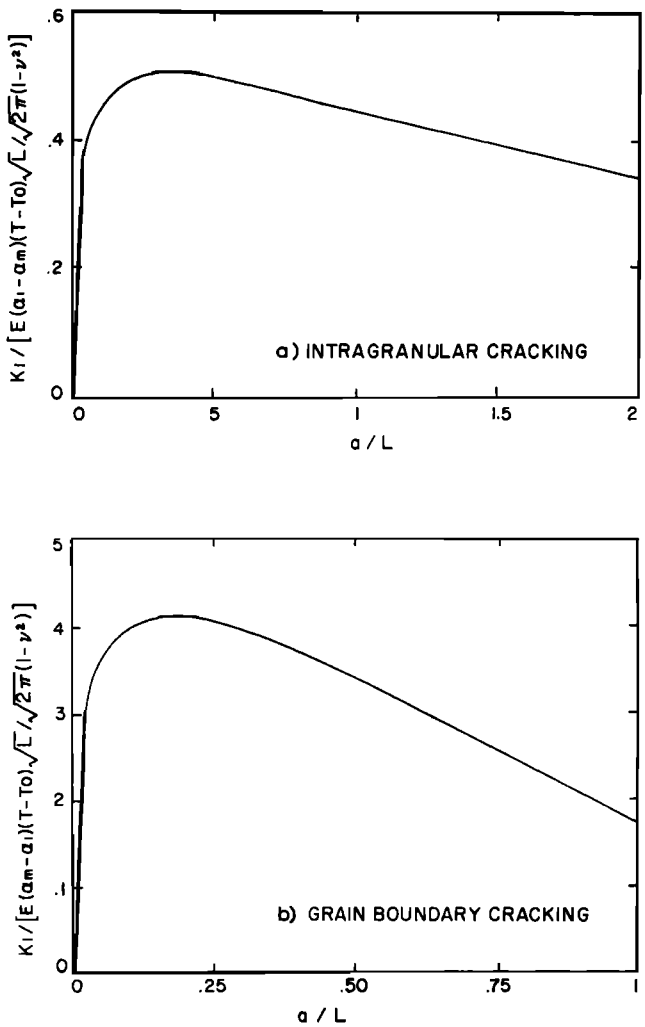



Fig. 14. Normalized stress intensity factor $K_1$ for cracking due to thermal expansion anisotropy in the square inclsion model plotted as a function of the normalized preexisting crack length a: (a) intragranular and (b) grain boundary cracking.

Our formalism is identical to that of Bruner [1984] who considered a more complicated inclusion geometry. However, our conclusion with regard to stablization and arrest behavior of intragranular cracking is drastically different from that of Bruner's study which predicted intragranular cracks to be generally unstable and to propagate across the grain. Although Bruner's concern was cracking induced by a temperature decrease during unroofing, the qualitative behavior should be identical to ours if we change the sign for the thermal expansion mismatch or anisotropy. A similar problem (of microcracking induced on cooling after the sintering process) is also treated in the ceramics literature. In fact, our approach was first suggested by Evans [1974] who considered thermal cracking due to thermal contraction hererogenity in ceramics. The key difference between our and Bruner's [1984] approach seems to be that we modeled intragranular cracking as microcracking emanating from the inclusion boundary into the matrix, whereas Bruner modeled the process as microcracking from the boundary into the inclusion.

It should be noted that Bruner's approach is expected to predict unstable crack growth for a variety of inclusion geometries. Eshelby's [1957] classic work has demonstrated that the stresses inside an ellipsoidal inclusion are homogeneous. A well-known result from fracture mechanics is that if the stress value in the integrand of (7) is constant, then $K_1$ so evaluated will increase monotonically as a function of crack length. The stress distribution is expected to be inhomogeneous inside an inclusion which is not ellipsoidal. However, it can be seen from published results involving other geometries (e. g., Chiu's [1977] work on a cuboidal inclusion and Bruner's results) that the variation of stresses within the inclusion is not very significant. Therefore it is not surprising that Bruner's results indicate an increase of $K_1$ with crack length.

We see no physical reason to choose one approach over the other. Our principal justification for adopting Evans' [1974] approach is simply that we observed little evidence for throughgoing intragranular cracking under the SEM or appreciable crack intersection and coalescence in our high-temperature samples (Figures 4-7 and 9).

The observations and analysis we presented here applies to microcracking induced by a homogeneous increase of temperature. The effects of a number of variables, notably thermal gradient, confining pressure, and pore fluid have not been considered. From a theoretical point of view the effects of thermal gradient on microcracking can probably be analyzed using thermoelasticity theory. If the elastic moduli heterogenity is relatively small, then the effect of pressure will simply be to reduce the stress intensity factor for a crack of length 2a by an amount $p(\pi a)^{1/2}$. The critical stress intensity factor is expected to be lower in a saturated rock than in a dry rock due to stress corrosion mechanisms causing subcritical crack growth. More experimental studies on the effects of heating rate, pressure, and pore fluids have to be conducted before we have a detailed understanding of the micromechanical processes responsible for thermal cracking in geological situations.

Westerly granite. The theoretical analyses imply that grain boundary cracking due to thermal expansion mismatch is expected to be unstable (Figure 12b), whereas that due to thermal expansion anisotropy can potentially be arrested (Figure 14b). Thermal expansion mismatch and anisotropy in Westerly granite are of the same order of magnitude (Appendix 2), and the net result is for most grain boundary cracks to propagate unstably with arrest only possibly occurring at geometric junctures with sharp changes in geometry. On the other hand, intragranular cracking due to both thermal expansion mismatch (Figure 12a) and anisotropy (Figure 14a) is expected to be relatively stable and crack growth therefore proceeds in incremental steps. The theoretical predictions are in general agreement with the SEM observations.

The theoretical results may also be used to predict the temperature at which thermally induced cracking will commence for a given inital microcrack length. Appropriate values for the thermoelastic moduli and critical stress intensity factor have to be chosen as input parameters for this calculation, the details of which are given in Appendix 2. We have chosen the physical parameters such that the theoretical results represent lower bounds on the crack propagation temperature, particularly for the case of intragranular cracking. Also note that we have performed the calculations only for the square inclusion model.

So far as the mean thermal expansion coefficients of the different phases of Westerly granite are concerned, the highest and lowest values are those of quartz and microcline respectively. Grain boundary cracking is therefore relatively easy at the interface between a microcline grain and another mineral. The stress intensity factor for cracks located at two of the four corners will also be enhanced by thermal expansion anisotropy (Figure 13d). We calculated the crack propagation temperature as a function of initial crack length at such critical corner locations under the combined influence of thermal expansion mismatch and anisotropy (Figure 15a). Similarly, the critical locations for intragranular cracking are expected to be at two of the corners outside a quartz grain such as shown in Figure 13c. We again calculated the crack propagation temperature for this case under the combined influence of thermal expansion mismatch and anisotropy (Figure 15a).

The longest cracks measured in virgin Westerly granite were less than 600 μm, and the arithmetic mean was about 18 μm [Hadley, 1976]. The range of temperatures we observed for grain boundary and intragranular cracking (Figures 2a and 2b) are in general agreement with those that one would expect on the basis of the virgin crack length spectra.

The SEM observations and fracture mechanics calculations provide insights on the interpreterion of acoustic emission data [Johnson et al., 1978; Chen and Wang, 1980]. It is generally observed that acoustic emissions can be detected at an initiation termperature of about 100°C for Westerly granite (although the exact value varies somewhat from system to system). The widest range of temperature investigated is probably that of Johnson et al. [1978] and their data clearly show a trend for the acoustic emission rate to increase drastically at about

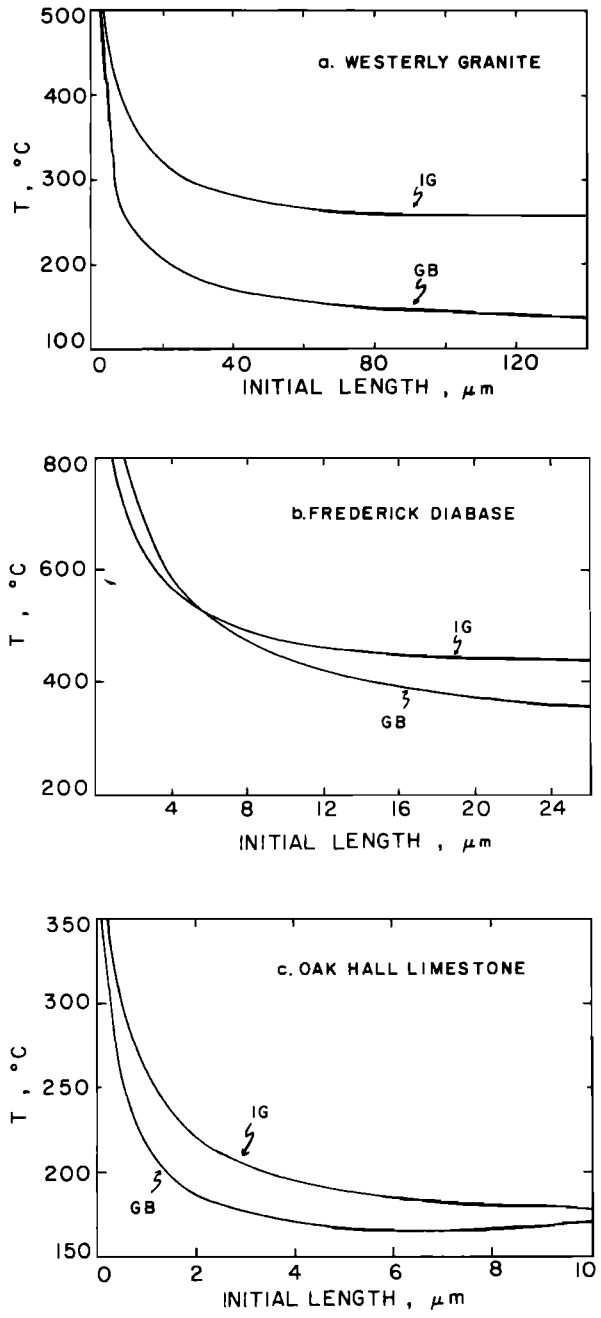



Fig. 15. Temperature for grain boundary (GB) and intragranular (IG) crack propagation induced by thermal expansion mismatch and anisotropy plotted as a function of initial crack length for the square inclusion model: (a) Westerly granite, (b) Frederick diabase, and (c) Oak Hall limestone.

300°C. This is comparable to the temperature at which our stereological data indicate a large increase in the rate of intragranular cracking (Figure 2b). Our theoretical calculations show that this mode of cracking proceeds in incremental steps and is therefore expected to generate acoustic emission signals which are relatively rich in high-frequency content.

Frederick diabase. Thermal expansion mismatch and anisotropy in the diabase are of the same order of magnitude, and we therefore expect grain boundary and intragranular crack propagation behavior similar to that discussed above for Westerly granite. We have also calculated the crack propagation temperature for a given crack length for Frederick diabase.

Using arguments similar to those discussed above for Westerly granite, we expect grain boundary cracking to be easiest at plagioclase grain boundaries and intragranular cracking to be most favorable at two of the corners of an augite grain (see Appendix 2). The results of the calculation are shown in Figure 15b. Note that for corresponding crack lengths, the crack propagation temperatures for Frederick diabase are generally about 200°C higher than those for Westerly granite. The reasons for this are twofold. First, the thermal expansion mismatch for the constituent minerals of the diabase is significantly less than that for the granite (Appendix 2). Second, the grain size of the diabase is about 4 times smaller than that of the granite (Table 1), and since the stress intensity factor $K_1$ scales as the product of the temperature increase and the square root of the grain size, a higher temperature is required for an aggregate of smaller grain size to attain the critical stress intensity factor necessary for cracking.

The theoretical predictions are in general agreement with our SEM observations. Thermal crack propagation was not observed in the diabase until after heating to 500°C. The corresponding temperature in the granite was 165°C. The negligible initial crack porosity of the diabase (Table 1) will also serve to limit cracking since there will be fewer cracks at which stress will become sufficiently concentrated so as to inititate propagation.

Oak Hall limestone. Although the thermal expansion anisotropy in the limestone is significant, the thermal expansion mismatch is zero. As a result, we expect both grain boundary and intragranular crack propagation to be stable and crack growth to therefore proceed in incremental steps (Figures 14a and 14b). We have again calculated the temperatures at which thermally induced crack propagation should occur for a given initial crack length. The results are shown in Figure 15c.

The grain size of the limestone is almost an order of magnitude smaller than that of the granite (Table 1). However, this stabilizing effect is counteracted by the greater thermal expansion anisotropy of the limestone (Appendix 2). The net result is that for corresponding crack lengths, the propagation temperature for the limestone is significantly lower than that for the granite. Note that although the crack porosity of the limestone is negligible (Table 1), crack growth should still be possible for cracks of initital length down to 1 μm after heating to temperatures greater than about 250°C.

As discussed previously, we observed essentially no thermal cracking in the limestone even after heating to 500°C. Thermal cracking has been reported in a variety of other calcitic rocks, all of which exhibit a grain size larger than that of Oak Hall limestone. These include the Carrara [Homand-Etienne and Troalen, 1984], Danby [Wong and Brace, 1979] and Yule [B. Evans, personal communication 1985] marbles, all of which have a grain size ranging from 200 to 300 μm. Thermally induced cracking has also been observed

in the Wombeyan marble [Rosengren and Jaeger, 1968] and Crepey oolitic limestone [Homand-Etienne and Troalen, 1984], both of which have a grain size of the order of millimeters.

A possible explanation for the discrepancy between SEM observation and the theoretical prediction is that the internal stresses induced by thermal expansion anisotropy have been relaxed by plastic flow processes operative in this relatively fine grained limestone. Therefore the linear elastic fracture mechanics concepts we have used to analyze thermal cracking may not be applicable. Plastic deformation induced by thermal expansion anisotropy was also reported by Boas and Honeycomb [1947] in their study of noncubic metals (which are anisotropic with respect to thermal expansion) subjected to heat treatment. Grain boundary migration was associated with the deformation (although they did not observe a grain size dependence).

Two types of mechanisms may be involved. If the preexisting microcracks in the limestone are predominately submicron in dimension, then brittle cracking requires a relatively high deviatoric stress exceeding several hundred megapascals (Figures 14a, 14b and 15c), below which processes such as twinning and crystal plasticity may have operated to relax the thermally induced internal stresses.

Another possibility is superplastic flow by diffusional creep and grain boundary sliding which is generally recognized as a grain size dependent mechanism of flow [Poirier, 1985]. Superplasticity has been observed in laboratory experiments involving the fine-grained (4 μm) Solnhofen limestone [Schmid et al., 1977] but was not observed in similar experiments on the coarser grained Carrara (200 μm) and Yule (300 μm) marbles [Schmid et al., 1980; Heard and Raleigh, 1972]. Preliminary calculations by us using Evans and Clarke's [1980] model indicate that elastic stress relaxation by grain size sensitive flow mechanisms in the Oak Hall limestone cannot be ruled out at the heating rate and temperature at which our experiments were performed. Further experiments and transmission electron microscopy studies need to be conducted to clarify this issue.

## Conclusion

We have studied in detail the micromechanics of thermally induced cracking in three common crustal rocks: a granite, a diabase and a limestone. Electron microscopy observation indicates that the thermally induced crack density is dependent upon the temperature, thermal expansion mismatch, thermal expansion anisotropy, initial crack porosity, and grain size. Grain boundary crack propagation is found to depend strongly on the local boundary geometry and to differ fundamentally from intragranular crack propagation. No appreciable crack coalescence is observed.

The thermally induced crack surface area per unit volume ($S_V$) in Westerly granite appears to have a quadratic dependence on the temperature increase. A physical explanation for such an empirical relation can be formulated on the basis of energetic balance. We have developed a square inclusion model to interpret our SEM observations. Fracture mechanics concepts are used to model the crack propagation and arrest behavior for grain boundary and intragranular cracks due to thermal expansion mismatch and anisotropy. The theoretical predictions for the crack propagation temperatures agree well with the stereological data for grain boundary and intragranular crack densities for Westerly granite. The predictions qualitatively match the observed behavior for Frederick diabase. Theoretical predictions for the fine-grained Oak Hall limestone do not agree with the observed behavior. It is suggested that the limestone may be deforming plastically and therefore the fracture mechanics concepts do not apply.

## Appendix 1

We will describe here the detailed derivation of the results used in the fracture mechanics analysis of thermal cracking. As discussed in the text, we modeled the internal stresses induced by thermal expansion heterogeneity as the stress field due to an inhomogeneous inclusion embedded in an effective medium. Once the stress field has been obtained for a given inclusion geometry and a given thermal expansion mismatch or anisotropy, the stress intensity factor for the tensile mode of cracking ($K_1$) was evaluated using a superposition technique.

### Evaluation of $K_1$ by the Superposition Technique

The values for the stress intensity factor shown in Figures 12a, 12b, 14a,and 14b were all evaluated using the so-called "superposition technique." Specifically, if a two-dimensional crack of length a is located along the x axis such that one tip is at the origin and the other at x=a (Figures 11b, 11c, 13c,and 13d) and if the tensile stress acting normal to the x axis in the absence of the crack has been determined to be $\sigma_{yy}(x)$, then the stress intensity factor $K_1$ at the tip x=a may be evaluated by the following integral:

$$K_1 = \left(\frac{2}{\pi a}\right)^{1/2} \int_0^a \frac{\sigma_{yy}(x)\, x^{1/2}}{(a-x)^{1/2}}\, dx \qquad (A1)$$

The use of this integral is motivated by the observation that the stress intensity factor due to the application of a pair of point forces (at x) on the surfaces of a crack (of length a) in an infinite medium is given by $(2/\pi a)^{1/2}[x/(a-x)]^{1/2}$ [Paris and Sih, 1965]. Equation (A1) therefore represents the superposition of this Green's function [Cartwright and Rooke, 1979].

### Stress Field Induced by Thermal Expansion Mismatch and Anisotropy: The Square Inclusion Case

The two-dimensional stress fields in the matrix surrounding a square inclusion resulting from thermal expansion mismatch (equations (3) and (5)) and thermal expansion anisotropy (equations (6) and (7)) were obtained by Eshelby's [1957] approach. If the matrix with the inclusion is subjected to a homogeneous increase in temperature (from $T_o$ to T), the inclusion will undergo what Eshelby termed a ''transformational strain'' ($e^T_{ij}$) which is given by $\Delta\alpha(T-T_o)\,\delta_{ij}$ for the

thermal expansion mismatch case (see equation (2a) and by $\Delta\alpha'_{ij}(T-T_o)$ for the thermal expansion anisotropy case (see equation (2b)).

A fundamental result obtained by Eshelby [1957] is that if $C_{ijkl}$ is the elastic moduli and $n_j$ is the outward normal to the inclusion boundary, then the stress field in the matrix (the "constrained stress") is simply that induced by a layer of body force of magnitude $C_{ijkl}e^T_{kl}n_j$ acting over the inclusion boundary. In an infinite, elastic space which is isotropic and homogeneous, the stress field due to a point force is given by Kelvin's solution, and hence the stress field in the matrix can be obtained by formally performing a surface integration of all the contributions from the layer of body force acting over the inclusion boundary.

Since we are only interested in a two-dimensional problem, we could have started with a two-dimensional Green's function and integrated all the contributions from point forces distributed over the square incluion boundary. This was the approach taken by Evans [1978] and Clarke [1980] who focused on the analysis of grain boundary cracking and therefore only obtained expressions for the normal stresses acting on the inclusion boundary. However, for the analysis of intragranular cracking, we also need to evaluate the stress field in the matrix in an area away from the inclusion. Since the three-dimensional problem for a cuboidal inclusion has been analysed thoroughly by Faivre [1964], Chiu [1977] and Lee and Johnson [1977], the solution to our two-dimensional problem for a square inclusion may be obtained from the three-dimensional results by evaluating the limit when one of the dimensions of the cuboid is infinitely long.

For the thermal expansion mismatch problem described by a "transformational strain" $e^T_{ij} = \Delta\alpha(T-T_o)\,\delta_{ij}$, we used Faivre's [1964] and Chiu's [1977] results to obtain, after some algebra, the stress field in the matrix (with reference to the coordinate system shown in Figure 14a):

$$\sigma_{xx} = -\sigma_{yy} = \frac{E\Delta\alpha(T-T_o)}{2\pi\,(1-\nu)}\left(\tan^{-1}\frac{y+L}{x+L} + \tan^{-1}\frac{y-L}{x-L} - \tan^{-1}\frac{y-L}{x+L} - \tan^{-1}\frac{y+L}{x-L}\right)$$

$$\sigma_{xy} = \frac{E\Delta\alpha(T-T_o)}{2\pi(1-\nu)}\ln\left[\frac{(x+L)^2 + (y+L)^2}{(x+L)^2+(y-L)^2}\cdot\frac{(x-L)^2 + (y-L)^2}{(x-L)^2 + (y+L)^2}\right] \tag{A2}$$

To evaluate the stress intensity factor using equation (A1), it is more convenient to express the stress tensor with respect to the two different coordinate systems shown in Figure 11b (for intragranular cracking) and Figure 11c (for grain boundary cracking). It is straightforward to show that after tensor transformation, the normal stresses $\sigma_{yy}$ will have the final form given of equations (3) and (5).

To illustrate the effects of thermal expansion anisotropy, we considered the case shown in Figure 13a. It was pointed out in the text that this case should be identical to that shown in Figure 13b after tensor transformation. This latter situation corresponds to a "transformational strain" $e^T_{12} = e^T_{21} = \Delta\alpha'(T-T_o)$. We used the three-dimensional results of Chiu [1977] and Lee and Johnson [1977] to derive the following stress field in the matrix (with reference to the coordinate system shown in Figure 13b):

$$\sigma_{xx} + \sigma_{yy} = \frac{E\Delta\alpha'(T-T_o)}{2\pi\,(1-\nu)}\ln\left[\frac{(x+L)^2 + (y+L)^2}{(x+L)^2 + (y-L)^2}\,\frac{(x-L)^2 + (y-L)^2}{(x-L)^2 + (y+L)^2}\right]$$

$$\sigma_{xx} - \sigma_{yy} = \frac{E\Delta\alpha'(T-T_o)}{\pi\,(1-\nu^2)}\left[\frac{(x+L)^2}{(x+L)^2 + (y+L)^2} + \frac{(x-L)^2}{(x-L)^2 + (y-L)^2} - \frac{(x+L)^2}{(x+L)^2 + (y-L)^2} - \frac{(x-L)^2}{(x-L)^2 + (y+L)^2}\right]$$

$$\sigma_{xy} = \frac{-E\Delta\alpha'(T-T_o)}{2\pi\,(1-\nu^2)}\left[\frac{(x+L)\,(y+L)}{(x+L)^2 + (y+L)^2} + \frac{(x-L)\,(y-L)}{(x-L)^2 + (y-L)^2} - \frac{(x+L)\,(y-L)}{(x+L)^2 + (y-L)^2} - \frac{(x-L)\,(y+L)}{(x-L)^2 + (y+L)^2}\right] \tag{A3}$$

To evaluate the stress intensity factor using (A1), it is more convenient to express the stresses with reference to the coordinate systems shown in Figure 13c (for intragranular cracking) and Figure 13d (for grain boundary cracking). It is straightforward to show that the expressions for the normal stress reduce to those given in equations (6) and (7) after tensor transformation.

## Appendix 2

To compare the theoretical predictions with the SEM observations, we need to choose appropriate values for the thermoelastic moduli and the fracture toughness. Since we focused on microcracking, it was appropriate to use measurements of the critical stress intensity factory $K_{1c}$ for single crystals. As discussed below, we consistently chose a low value for $K_{1c}$. Therefore the values that we used are definitely lower bounds on the fracture toughness corresponding to intragranular cracking. We are not aware of any systematic measurements of $K_{1c}$ for grain boundary cracks, although it is generally assumed that they should be lower than

those for single crystals. In this sense, the low values of $K_{1c}$ that we chose are probably representative for grain boundary cracking.

For the thermal expansion coefficients of the matrix, we chose measurements on the polycrystalline rock from 25°C to 200°C at room pressure obtained by Richter and Simmons [1974]. The thermal expansivity tensors for the inclusions were chosen to be those of the various minerals measured from 20°C to 200°C and compiled by Skinner [1966].

Both the Young's modulus and Poisson's ratio are expected to be dependent on stress [Brace, 1971] and temperature [Johnson et al., 1978]. We used values of ultrasonic velocities determined from measurements of the resonant frequency of bars. Room temperture ($T_o$) was taken to be 20°C.

The dimension of the square (2L) was chosen as the grain size. We have consistently used the values published by Brace [1965] and given in Table 1 for our calculations. Our data for $P_L$ for grain boundaries (Table 3) for Westerly granite may also be used to calculate the grain size for this rock, given the initial assumption that all grains are the same size and shape and are space filling [see Underwood, 1970, chapter 4]. Depending on the geometry chosen, grain sizes range from 0.5 mm (for a sphere) to 0.6 mm (for an octahedron) to 1.3 mm (for a tetrahedron). The grain sizes of 0.75 mm used here is a reasonable average of the many possible values. In addition, note that since the stress intensity factor $K_1$ scales as the square root of the grain size, the calculation is not very sensitive to small changes in the grain dimension.

Westerly Granite

$K_{1c}$ = 0.3 MPa $m^{1/2}$ (lowest value for quartz, microcline and plagioclase reported by Atkinson [1984]
E = 46 GPa (average of the values from 20°C to 200°C measured by Johnson et al. [1978]
v = 0.1 [Brace, 1971]
$\alpha^o$ = 11.2 X $10^{-6}$/°C [Richter and Simmons, 1974]
L = 0.375 mm

1. Grain boundary cracking: microline ($Or_{83}Ab_{17}$) inclusion

$\alpha_{11}$ = 16.3 X $10^{-6}$/°C
$\alpha_{22}$ = 0.2 X $10^{-6}$/°C
$\alpha_{33}$ = 0.6 X $10^{-6}$/°C [Skinner, 1966]
$\Delta\alpha$ = -5.5 X $10^{-6}$/°C
$\Delta\alpha'$ = 8 X $10^{-6}$/°C

2. Intragranular cracking: quartz inclusion

$\alpha_{11} = \alpha_{22}$ = 16.7 X $10^{-6}$/°C
$\alpha_{33}$ = 10 X $10^{-6}$/°C [Skinner, 1966]
$\Delta\alpha$ = 3.3 X $10^{-6}$/°C
$\Delta\alpha'$ = 3.4 X $10^{-6}$/°C

Frederick Diabase

$K_{1c}$ = 0.3 MPa $m^{1/2}$ (lowest value for plagioclase reported by Atkinson [1984]; no data for pyroxene)
E = 100 GPa (room temperature value measured by Brace [1964])
v = 0.23 [Simmons and Brace, 1965]
$\alpha^o$ = 6.3 X $10^{-6}$/°C [Richter and Simmons, 1974]
L = 0.088 mm

1. Grain boundary cracking: plagioclase ($Ab_{56}An_{44}$) inclusion

$\alpha_{11}$ = 6.1 X $10^{-6}$/°C
$\alpha_{22}$ = 3.3 X $10^{-6}$/°C
$\alpha_{33}$ = 3.9 X $10^{-6}$/°C [Skinner, 1966]
$\Delta\alpha$ = -1.9 X $10^{-6}$/°C
$\Delta\alpha'$ = 1.4 X $10^{-6}$/°C

2. Intragranular cracking: augite inclusion

$\alpha_{11}$ = 3.3 X $10^{-6}$/°C
$\alpha_{22}$ = 10.6 X $10^{-6}$/°C
$\alpha_{33}$ = 5.6 X $10^{-6}$/°C [Skinner, 1966]
$\Delta\alpha$ = 0.18 X $10^{-6}$/°C
$\Delta\alpha'$ = 3.6 X $10^{-6}$/°C

Oak Hall Limestone

$K_{1c}$ = 0.2 MPa $m^{1/2}$ (lowest value for calcite reported by Atkinson [1984])
E = 79 GPa
v = 0.28 [Simmons and Brace, 1965]
L = 0.04 mm

1. Grain boundary cracking

$\alpha_{11} = \alpha_{22}$ = -5.3 X $10^{-6}$/°C
$\alpha_{33}$ = 26.4 X $10^{-6}$/°C [Skinner, 1966]
$\Delta\alpha$ = 0
$\Delta\alpha'$ = 15.8 X $10^{-6}$/°C

2. Intragranular cracking; same as grain boundary cracking.

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